Generalization of semi-regular sequences: Maximal Gröbner basis degree, variants of genericness, and related conjectures*

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Abstract

Nowadays, the notion of semi-regular sequences, originally proposed by Fröberg, becomes very important not only in Mathematics, but also in Information Science, in particular Cryptology. For example, it is highly expected that randomly generated polynomials form a semi-regular sequence, and based on this observation, secure cryptosystems based on polynomial systems can be devised. In this paper, we deal with a semi-regular sequence and its variant, named a generalized cryptographic semi-regular sequence, and give precise analysis on the complexity of computing a Gröbner basis of the ideal generated by such a sequence with help of several regularities of the ideal related to Lazard's bound on maximal Gröbner basis degree and other bounds. We also study the genericness of the property that a sequence is semi-regular, and its variants related to Fröberg's conjecture. Moreover, we discuss on the genericness of another important property that the initial ideal is weakly reverse lexicographic, related to Moreno-Socías' conjecture, and show some criteria to examine whether both Fröberg's conjecture and Moreno-Socías' one hold at the same time.

1 Introduction

In the theory of polynomial rings over fields, Gröbner bases introduced first by Buchberger [9] provide various methods for investigating important properties of

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an ideal and its associated algebraic variety. Here, a Gröbner basis is defined as a special kind of its generator set, relative to a monomial ordering. Buchberger also presented the first algorithm (Buchberger's algorithm) for computing Gröbner bases. Lazard's formulation in [38] and [39] relates the problem of computing Gröbner bases to the problem of reducing so-called Macaulay matrices. FGLM [22] and Gröbner walk [13] were proposed as algorithms that efficiently convert a given Gröbner basis to a Gröbner basis with respect to another monomial ordering; Hilbert driven algorithm [46] can be also applied to such a conversion. Faugère proposed his innovative algorithms F_4 [20] and F_5 [21] allowing one to quite efficiently compute Gröbner bases, and these are nowadays state-of-the-art algorithms. We refer to [19] for a survey on signature-based algorithms including F_5. Thanks to Gröbner basis conversion algorithms, it is important to estimate the complexity of computing a Gröbner basis with respect to a specific monomial ordering. (Throughout this paper, all the complexities are measured by the number of arithmetic operations in a field to which all the coefficients of the input polynomials belong.) In this paper, we consider only the case of *graded* monomial orderings; in particular, we adopt a graded reverse lexicographic ordering.

In order to estimate the complexity, it suffices to provide an upper-bound on the so-called *solving degree*: Roughly speaking, the solving degree is the highest degree of polynomials involved in Gröbner basis computation, see [36, Section 2.2] for rigorous definitions. If the given ideal is homogeneous, the solving degree can be equal to the maximal degree of the reduced Gröbner basis with respect to a monomial ordering that one considers, and its upper-bounds have been well-studied. In the case of inhomogeneous ideals, the solving degree can be *greater than* the maximal degree of the reduced Gröbner basis, whence estimating it *directly* is difficult in general, see [11] and [36] for details. On the other hand, it is well-known (cf. [34, Prop. 4.3.18]) that a Gröbner basis of an inhomogeneous ideal with respect to a graded monomial ordering is easily obtained from a Gröbner basis of the ideal of *homogenized* generators. Therefore, the homogeneous case is essential, and we focus on that case. In the following, let K be a field, and let $R = K[x_1, \ldots, x_n]$ be the polynomial ring of n variables over K. The term "degree" means total degree, unless otherwise noted. Let R_d be its homogeneous part of degree d of R (i.e., the K-linear space spanned by the monomials in R of degree d) for each non-negative integer d. For an ideal (possibly inhomogeneous) I of R, we denote by $\mathrm{max.GB.deg}_{\prec}(I)$ the maximal degree of the elements of the reduced Gröbner basis of I with respect to a monomial ordering $\prec$ on the set $\mathcal{T}$ of monomials in R, following [10, Definition 7] (it is also denoted by $\deg(I, \prec)$ in the literature of commutative algebra, see e.g., [31, Definition 4]). If I is generated by $F \subset R$, we may write $\mathrm{max.GB.deg}_{\prec}(F)$ in place of it.

Lazard's bound and Hashemi-Seiler's inequality

For a *homogeneous* ideal $I \subset R = K[x_1, \ldots, x_n]$ with a generating set $F = \{f_1, \ldots, f_m\} \subset R$, it is well-known (see e.g., [3, Proposition 1], [45, Equation (24)]) that a Gröbner basis of I with respect to a graded monomial ordering up to degree d can be computed in $O(md\binom{n+d-1}{d}^{\omega})$, where $2 \leq \omega < 3$ is the exponent of matrix multiplication. (We can also obtain another upper-bound $O(m\binom{n+d}{d}^{\omega})$ by the proof of [44, Theorem 1.72].) From this fact, it follows in the *homogeneous* case that the complexity of computing the reduced Gröbner basis of I is upper-bounded by $O(mD\binom{n+D-1}{D}^{\omega})$ (or $O(m\binom{n+D}{D}^{\omega})$) for any integer D with $\mathrm{max.GB.deg}_{\prec}(I) \leq D$. Hence, finding a tighter bound D of $\mathrm{max.GB.deg}_{\prec}(I)$ is important for estimating shapely the complexity of Gröbner basis computation in the homogeneous case. The case on which we focus is the case where the Krull dimension of R/I, denoted by $\mathrm{Krull.dim}(R/I)$, is equal to or less than one; see e.g., [31, Section 1] for a summary of existing results when $\mathrm{Krull.dim}(R/I) \geq 2$. In the case where $\mathrm{Krull.dim}(R/I) \leq 1$, Lazard's bound (called also Macaulay's bound) is well-known. Specifically, if K is large enough, then there exists a (suitable) linear coordinate change σ such that $R/(I^{\sigma} + \langle x_n \rangle)$ is Artinian (equivalently I is in *quasi stable position*, see Lemma 2.1.8 below). From this, we assume without loss of generality that $R/(I + \langle x_n \rangle)$ is Artinian. In this case, putting $d_j = \deg(f_j)$ for $1 \leq j \leq m$ and supposing that $d_1 \geq d_2 \geq \cdots \geq d_m$ (descending order), it follows that

$$\mathrm{max.GB.deg}_{\prec}(I) \leq \sum_{j=1}^{\min\{m,n\}} (d_j - 1) + 1 \tag{1.1}$$

for the graded reverse lexicographical ordering $\prec$ with $x_n \prec \cdots \prec x_1$, see [39, Théorème 3.3] and [38, Theorem 2]. Recently, Hashemi-Seiler [31] re-proved Lazard's bound by using the inequality

$$\mathrm{max.GB.deg}_{\prec}(I) \leq \max\{\mathrm{hilb}(I + \langle x_n \rangle), \mathrm{hilb}(I)\}, \tag{1.2}$$

where $\mathrm{hilb}(J)$ denotes the Hilbert regularity of a homogeneous ideal J of R. Note that $\mathrm{hilb}(I + \langle x_n \rangle)$ is equal to the *degree of regularity* $d_{\mathrm{reg}}(I + \langle x_n \rangle)$ since $R/(I + \langle x_n \rangle)$ is Artinian, and $\mathrm{hilb}(I)$ is equal to the *generalized degree of regularity* $\widetilde{d}_{\mathrm{reg}}(I)$. For the definition of d_{reg} and $\widetilde{d}_{\mathrm{reg}}$, see Remark 2.1.2 below.

Remark 1. Note that the condition $\mathrm{Krull.dim}(R/I) \leq 1$ does not necessarily imply $d_{\mathrm{reg}}(I + \langle x_n \rangle) = \mathrm{hilb}(I + \langle x_n \rangle) < \infty$, without considering any linear coordinate change σ. For example, putting $F := \{f_1 := x_1x_2 - 1, f_2 := x_2 - 1\} \subset K[x_1, x_2]$ (inhomogeneous), it is obvious that $\mathrm{Krull.dim}(R[y]/I) = 1$ for $I = \langle F^h \rangle$ but $d_{\mathrm{reg}}(I + \langle y \rangle) = d_{\mathrm{reg}}(\langle F^{\mathrm{top}} \rangle) = \infty$, where F^h is the homogenization of F by an extra variable y, and where F^{top} is the homogeneous part of highest degree, see Remark 3 below. In this case, $F^h = \{x_1x_2 - y^2, x_2 - y\}$ and $F^{\mathrm{top}} = \{x_1x_2, x_2\}$.

On the other hand, if K is a finite field of order q and if we would compute a zero of $\langle F \rangle$ over K, we often add the set of so-called field equations $x_i^q - x_i$ to an inhomogeneous F, say $F_1 := F \cup \{x_i^q - x_i : 1 \leq i \leq n\}$, so that we always have $d_{\mathrm{reg}}(\langle F_1^h \rangle + \langle y \rangle) = d_{\mathrm{reg}}(\langle F_1^{\mathrm{top}} \rangle) < \infty$.

Our results and conjectures

In this paper, we focus on several cases that are expected to be *generic* in some sense, and obtain upper-bounds on $\mathrm{max.GB.deg}_{\prec}(I)$ and some results related to Fröberg's conjecture [23] and Moreno-Socías' one [41], in those cases. For example, we will assume that $R/(I + \langle x_n \rangle)$ has the Hilbert series associated to generic forms, as conjectured by Fröberg [23] (see Conjecture 2.3.7 below). Note that a homogeneous polynomial sequence is called *cryptographic semi-regular* (cf. [8]) if its associated coordinate ring has the Hilbert series conjectured by Fröberg, see Definition 2.2.3 and Proposition 2.2.4 below for details. In the literature, a cryptographic semi-regular sequence plays a very important role in analyzing Gröbner basis computation, both in theory and in practical applications to cryptography, see e.g., Bardet et al's works [1], [2], [3], [4], Gorla et al.'s works [8], [10], [11], and our ones [35], [36]. In our earlier work [36], when $m \geq n$, we improved Lazard's bound on $\mathrm{max.GB.deg}_{\prec}(F)$ to $\sum_{j=1}^{n}(d_j - 1) + 1$ with $d_1 \leq \cdots \leq d_m$ (ascending order), supposing that $(f_1|_{x_n=0}, \ldots, f_i|_{x_n=0})$ is cryptographic semi-regular on $K[x_1, \ldots, x_{n-1}]$ for each $1 \leq i \leq m$. In [36], we also defined a *generalized cryptographic semi-regular* sequence (see Definition 2.2.8 below for the definition and Remark 2.2.9 for a construction), as an extension of cryptographic semi-regular sequence to the case of Krull dimension one.

As the first result of this paper, we prove Theorem 1.1, which provides an upper-bound on $\mathrm{max.GB.deg}_{\prec}(I)$ for a generalized cryptographic semi-regular sequence and a complexity bound on computing a Gröbner basis of I.

Theorem 1.1. *Let $R = K[x_1, \ldots, x_n]$ be the polynomial ring of n variables over a field K (in any characteristic), and $\prec$ the graded reverse lexicographical ordering with $x_n \prec \cdots \prec x_1$. Let I be a homogeneous ideal of R generated by homogeneous polynomials $f_1, \ldots, f_m$ in $R \smallsetminus K$ such that* $\mathrm{Krull.dim}(R/I) \leq 1$. *Assume that $(f_1|_{x_n=0}, \ldots, f_m|_{x_n=0})$ is cryptographic semi-regular on $K[x_1, \ldots, x_{n-1}]$ (equivalently $(f_1, \ldots, f_m, x_n)$ is cryptographic semi-regular), and that $(f_1, \ldots, f_m)$ is generalized cryptographic semi-regular, Then the following (1) and (2) hold:*

(1) We have

$$\mathrm{max.GB.deg}_{\prec}(I) \leq \max\{d_{\mathrm{reg}}(I + \langle x_n \rangle), \widetilde{d}_{\mathrm{reg}}(I)\} \leq D^{(n,m)}, \tag{1.3}$$

where

$$D^{(n,m)} := \begin{cases} \deg\left(\left[\frac{\prod_{i=1}^{m}(1-z^{d_i})}{(1-z)^n}\right]\right)+1 & (m \geq n), \\ \sum_{i=1}^{n-1}(d_i-1)+1 & (m = n-1). \end{cases} \tag{1.4}$$

(2) *The complexity of computing a Gröbner basis of I with respect to $\prec$ is upper-bounded by*

$$O\left(m\binom{n+D^{(n,m)}-1}{D^{(n,m)}}^{\omega}\right) \tag{1.5}$$

arithmetic operations in K, where $2 \leq \omega < 3$ is the exponent of matrix multiplication.

The bound $D^{(n,m)}$ in (1.3) is optimal for a homogeneous ideal I satisfying the assumptions of Theorem 1.1, see Example 3.1.4 below. The proof of (2) in Theorem 1.1 is based on a well-known fact that the a Gröbner basis of I is obtained by computing the reduced row echelon form of the degree-D Macaulay matrix for any D with $D \geq \mathrm{max.GB.deg}_{\prec}(I)$, see Section 3.1 for details.

Our assumption that a homogeneous polynomial sequence is generalized cryptographic semi-regular could be useful to analyze the security of multivariate cryptosystems and so on (cf. [36, Section 4.3]). The bound $D^{(n,m)}$ given in (1.4) coincides with Lazard's bound when $m \in \{n-1, n\}$, while it can be sharper than Lazard's one when $m > n$. We also note that, the complexity bound (1.5) is slightly better than the usual bound $O(mD\binom{n+D-1}{D}^{\omega})$ (see e.g., [3, Proposition 1], [45, Equation (24)]) and than $O(m\binom{n+D}{D}^{\omega})$ (which comes from the proof of [44, Theorem 1.72]) for an arbitrary D with $D \geq \mathrm{max.GB.deg}_{\prec}(F)$.

As the second result, we shall prove that Hashemi-Seiler's inequality (1.2) is as possible as tight under some assumptions, one of which is that an initial ideal is *weakly reverse lexicographic*. Here, a weakly reverse lexicographic ideal is a monomial ideal J such that if x^{α} is one of the minimal generators of J then every monomial of the same degree which preceeds x^{α} must belong to J as well. Moreno-Socías [41] conjectured that, for the graded reverse lexicographical ordering, the initial ideal of a generic ideal is a weakly reverse lexicographic.

Theorem 1.2. *Let $R = K[x_1, \ldots, x_n]$ be the polynomial ring of n variables over a field K (in any characteristic), and let $\prec$ be the graded reverse lexicographical ordering $\prec$ on R with $x_n \prec \cdots \prec x_1$. Let I be a proper homogeneous ideal of R with* $\mathrm{Krull.dim}(R/I) \leq 1$ *and* $\mathrm{Krull.dim}(R/(I + \langle x_n \rangle)) = 0$. *If $(f_1|_{x_n=0}, \ldots, f_m|_{x_n=0})$ is cryptographic semi-regular on $K[x_1, \ldots, x_{n-1}]$ (equivalently $(f_1, \ldots, f_m, x_n)$ is cryptographic semi-regular) and if the initial ideal* $\mathrm{in}_{\prec}(I) = \langle \mathrm{LM}_{\prec}(I) \rangle_R$ *is weakly reverse lexicographic, then the inequality (1.2) becomes an equality.*

After proving Theorems 1.1 and 1.2, in Section 4, we also study the genericness of our assumptions used in the theorems, based on the theory of *parametric Gröbner bases*. For this, in Section 2.3, we formulate the genericness of cryptographic semi-regular sequences in terms of parametric Gröbner bases. Then, in Section 4, some analogous results for generalized cryptographic semi-regular sequences will be proved, and the following conjectures will be raised as variants of Fröberg's conjecture:

Conjecture 1.3 (Conjecture 4.1.10; see also [36, Conjecture 4.3.4])**.** *Assume that K is an infinite field, and we endow $V_{n,m,(d_1,\ldots,d_m)} := R_{d_1} \times \cdots \times R_{d_m}$ naturally with the Zariski topology. For any n, m, and $(d_1, \ldots, d_m)$ and for any $\boldsymbol{p} \in \mathbb{P}^{n-1}(K)$, the property that $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ is generalized cryptographic semi-regular is generic on $Z_{\boldsymbol{p}} := \{\boldsymbol{H} \in V_{n,m,(d_1,\ldots,d_m)} : \boldsymbol{p} \in V_{\overline{K}}(\boldsymbol{H})\}$, namely there exists a non-empty Zariski-open set of $Z_{\boldsymbol{p}}$ with respect to the induced topology on which the property holds.*

Conjecture 1.4 (Conjecture 4.1.11)**.** *Assume that K is an infinite field. For any n, m, and $(d_1, \ldots, d_m)$, the property that $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ is generalized cryptographic semi-regular is generic on $V^{\dim \geq 1}_{n,m,(d_1,\ldots,d_m)} := \{\boldsymbol{H} \in V_{n,m,(d_1,\ldots,d_m)} : \mathrm{Krull.dim}(R/\langle \boldsymbol{H} \rangle) \geq 1\}$, namely there exists a non-empty Zariski-open set of $V^{\dim \geq 1}_{n,m,(d_1,\ldots,d_m)}$ with respect to the induced topology on which the property holds.*

We also show a new criterion to examine whether Fröberg's conjecture and Moreno-Socías' one hold at the same time, for a *fixed* parameter set $(n, m, d_1, \ldots, d_m)$ (although these conjectures are set for *any* parameter set, see Remark 2 for details):

Theorem 1.5 (Theorem 4.2.1)**.** *Assume that K is an infinite field, and let $n, m, d_1, \ldots, d_m$ be positive integers. If there exists a cryptographic semi-regular sequence $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ such that $\mathrm{in}_{\prec}(\langle \boldsymbol{F} \rangle)$ is weakly reverse lexicographic, then there also exists a non-empty Zariski-open set U of $V_{n,m,(d_1,\ldots,d_m)}$ such that $\boldsymbol{H}$ is cryptographic semi-regular and $\mathrm{in}_{\prec}(\langle \boldsymbol{H} \rangle)$ is weakly reverse lexicographic for any $\boldsymbol{H} \in U$. Namely, Fröberg's conjecture and Moreno-Socías' one both hold for the parameter set $(n, m, d_1, \ldots, d_m)$.*

Remark 2. Pardue proved in his seminal paper [43] that Fröberg's conjecture is equivalent to his several conjectures, and that Moreno-Socías' conjecture implies all of them. Note that these equivalence and implication are intended to mean that each conjecture holds for *any* choice of parameters $(n, m, d_1, \ldots, d_m)$. More precisely, we see from the proof of his theorem [43, Theorem 2] that (1) and (2) are equivalent to each other and that (3) implies (1) and (2), over an infinite field:

(1) (Fröberg's conjecture) The property that $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ is cryptographic semi-regular is generic (in terms of Definition 2.3.1) for any $(n, m, d_1, \ldots, d_m)$.

(2) (Pardue's conjecture [43, Conjecture B]) The property that $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ is semi-regular is generic for any $(n, m, d_1, \ldots, d_m)$.

(3) (Moreno-Socías' conjecture [41]) The property that the initial ideal of the homogeneous ideal generated by $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ with respect to a graded reverse lexicographical ordering is weakly reverse lexicographic is generic for any $(n, m, d_1, \ldots, d_m)$.

Namely, even if the property in (3) is shown to be generic for a fixed parameter set $(n, m, d_1, \ldots, d_m)$, it is unclear that the property in (1) (or (2)) is generic for the same parameter set (this also applies to the case of (1) $\Rightarrow$ (2)).

Analogously to Theorem 1.5, we finally prove the following theorem, which can also examine Conjecture 1.3:

Theorem 1.6 (Proposition 4.2.3 and Proposition 4.2.4)**.** *Under the same setting as in Theorem 1.5, assume that there exists a generalized cryptographic semi-regular sequence $\boldsymbol{F}$ in $V_{n,m,(d_1,\ldots,d_m)}$ with* $\mathrm{Krull.dim}(R/\langle \boldsymbol{F} \rangle) = 1$ *such that* $\dim_K(R/\langle \boldsymbol{F} \rangle)_D = 1$ *for* $D = \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{F} \rangle)$ *and* $\mathrm{in}(\langle \boldsymbol{F} \rangle)$ *is weakly reverse lexicographic. Then, for any* $\boldsymbol{p} \in \mathbb{P}^{n-1}(K)$*, there exists a non-empty Zariski-open set* $U_{\boldsymbol{p}}$ *of* $Z_{\boldsymbol{p}} := \{\boldsymbol{H} \in V_{n,m,(d_1,\ldots,d_m)} : \boldsymbol{p} \in V_{\overline{K}}(\boldsymbol{H})\}$ *with respect to the induced topology such that* $\boldsymbol{H}$ *is generalized cryptographic semi-regular and* $\mathrm{in}(\langle \boldsymbol{H} \rangle)$ *is weakly reverse lexicographic for any* $\boldsymbol{H}$ *in* $U_{\boldsymbol{p}}$*.*

To the end of Introduction, we remark an application of Theorem 1.1 to the inhomogeneous case.

Remark 3 (Inhomogeneous case)**.** By Theorem 1.1, we can also estimate the complexity of computing a Gröbner basis for an inhomogeneous polynomial sequence defining a zero-dimensional ideal, as in [36, Section 4.3]. Indeed, for a subset $F = \{f_1, \ldots, f_m\}$ of $R = K[x_1, \ldots, x_n]$ containing at least one inhomogeneous polynomial, a Gröbner basis of $\langle F \rangle_R$ is obtained by reducing the associated degree-D Macaulay matrix at $D = \mathrm{max.GB.deg}_{\prec^h}(I)$ with $I = \langle f_1^h, \ldots, f_m^h \rangle \subset R' := R[y]$, see [10, Theorem 7]. Here, each f_j^h denotes the homogenization of f_j by an extra variable y, and $\prec^h$ is the homogenization of $\prec$, see e.g., [34, Section 3.1.1] for definition. The complexity of reducing the Macaulay matrix can be easily estimated as $O(m\binom{n+D}{D}^{\omega})$, without assuming $D = O(n)$. Suppose the following three conditions:

- The Krull dimension of $R'/\langle f_1^h, \ldots, f_m^h, y \rangle$ is zero, namely the Krull dimension of $R/\langle f_1^{\mathrm{top}}, \ldots, f_m^{\mathrm{top}} \rangle$ is zero, where $f_j^{\mathrm{top}} := (f_j^h)|_{y=0}$ for $1 \leq j \leq m$. (Equivalently these two graded rings are both Artinian.)
- The sequence $(f_1^h, \ldots, f_m^h, y) \in (R')^{m+1}$ is cryptographic semi-regular, equivalently $(f_1^{\mathrm{top}}, \ldots, f_m^{\mathrm{top}}) \in R^m$ is cryptographic semi-regular.

- The sequence $(f_1^h, \ldots, f_m^h) \in (R')^m$ is generalized cryptographic semi-regular.

Then, it follows from Theorem 1.1 that $\max\{d_{\mathrm{reg}}(\langle F^{\mathrm{top}}\rangle), \widetilde{d}_{\mathrm{reg}}(\langle F^h\rangle)\} \leq D^{(n+1,m)}$ with $F^{\mathrm{top}} := \{f_1^{\mathrm{top}}, \ldots, f_m^{\mathrm{top}}\}$ and we have the complexity bound

$$O\left(m\binom{n + D^{(n+1,m)}}{D^{(n+1,m)}}^{\omega}\right). \tag{1.6}$$

Note that this upper-bound is heuristically found in [28].

2 Preliminaries

Let K be a field, and let $\overline{K}$ denote its algebraic closure. Let n be a natural number, and let $R = K[x_1, \ldots, x_n]$ be the polynomial ring of n variables over K. We denote by $\mathcal{T}$ the set of monomials in $\{x_1, \ldots, x_n\}$. As our convention, as graded structure objects, for a subset S of R and a non-negative integer d, we also denote by S_d the subset consisting of all elements of degree d in S. As an exception, if S is an ideal of R, it is allowed that S_d contains 0. The Krull dimension of a commutative ring A is denoted by $\mathrm{Krull.dim}(A)$. For ideals I and J of R, their ideal quotient is defined as $(I : J) = \{f \in R \mid fJ \subset I\}$. The saturation of I with respect to J is $(I : J^\infty) := \{f \in R \mid fJ^s \subset I \text{ for some } s \geq 1\}$. In particular, the saturation of a homogeneous ideal I with respect to the maximal homogeneous ideal $\mathfrak{m} = \langle x_1, \ldots, x_n\rangle_R$ is denoted by $I^{\mathrm{sat}} = (I : \mathfrak{m}^\infty)$, which is also homogeneous.

2.1 Hilbert regularity

Let I be a non-zero homogeneous ideal of R. Let $\mathrm{HP}_{R/I} \in \mathbb{Q}[z]$ denote the Hilbert polynomial of R/I, namely there exists a non-negative integer d_0 such that $\mathrm{HF}_{R/I}(d) = \mathrm{HP}_{R/I}(d)$ for any d with $d \geq d_0$, where HF is the Hilbert function.

Definition 2.1.1 (Hilbert regularity (or index of regularity [14, Chapter IX, Section 3])). The smallest d_0 such that $\mathrm{HF}_{R/I}(d) = \mathrm{HP}_{R/I}(d)$ for any d with $d \geq d_0$ is called the *Hilbert regularity* of I, denoted by $\mathrm{hilb}(I)$, or is called the *index of regularity* of I, denoted by $i_{\mathrm{reg}}(I)$.

Throughout the rest of this subsection, we set $r := \mathrm{Krull.dim}(R/I)$.

Remark 2.1.2. It is well-known that the following conditions are all equivalent:

(1) The Krull dimension r of R/I is equal to 1 (resp. 0).

(2) The projective algebraic set $V_{\overline{K}}(I)$ is a finite set (resp. an empty set).

(3) The Hilbert polynomial $\mathrm{HP}_{R/I}$ is a positive constant (resp. the zero).

(4) There exists a non-negative integer d_0 such that $\mathrm{HF}_{R/I}(d) = \mathrm{HF}_{R/I}(d_0) > 0$ (resp. $\mathrm{HF}_{R/I}(d) = 0$) for any d with $d \geq d_0$.

In this case, it is straightforward that

$$\begin{aligned} \mathrm{hilb}(I) &= \min\{d_0 : \mathrm{HF}_{R/I}(d) = \mathrm{HF}_{R/I}(d_0) \text{ for any } d \text{ with } d \geq d_0\} \\ &= \min\{d_0 : \dim_K(R/I)_d = \dim_K(R/I)_{d_0} \text{ for any } d \text{ with } d \geq d_0\}. \end{aligned}$$

The middle and the right hand sides are called the *generalized degree of regularity* of I for $r \leq 1$, denoted by $\widetilde{d}_{\mathrm{reg}}(I)$ (cf. [36, Sect. 2.3]): When $r = 0$, these are equal to the *degree of regularity* $d_{\mathrm{reg}}(I) := \min\{d : R_d = I_d\}$. Namely, we have

$$\mathrm{hilb}(I) = \begin{cases} \widetilde{d}_{\mathrm{reg}}(I) & (r = 1), \\ \widetilde{d}_{\mathrm{reg}}(I) \quad = d_{\mathrm{reg}}(I) & (r = 0). \end{cases} \tag{2.1.1}$$

See [8] for bounds on $d_{\mathrm{reg}}(I)$.

Although we mainly treat the case where $r \leq 1$ in this paper, we distinguish $\mathrm{hil}(I)$ and $\widetilde{d}_{\mathrm{reg}}(I)$ for the compatibility with the condition that a sequence of homogeneous polynomials in R is regular (cf. Lemma 2.2.12 below).

Remark 2.1.3. Note also that the Hilbert series $\mathrm{HS}_{R/I}$ is written as $\mathrm{HS}_{R/I}(z) = h_{R/I}(z)/(1-z)^r$ for some polynomial $h_{R/I}(z) \in \mathbb{Z}[z]$, which is called the *$h$-polynomial* of R/I. Then, we have $\mathrm{hilb}(I) = \deg h_{R/I} - r + 1$, see [7, Prop. 4.1.12].

Moreover, it follows from Grothendieck-Serre's formula that

$$\mathrm{hilb}(I) \leq \mathrm{reg}(I) = \mathrm{reg}(R/I) + 1,$$

where 'reg' means *Castelnuovo-Mumford regularity*, see [17, §20.5] for the definition. The equality holds if R/I is Artinian, i.e., the Krull dimension of R/I is zero, see e.g., [18, Corollary 4.4].

Definition 2.1.4 (cf. [31, Section 3])**.** We say that I is in *Noether position* with respect to the variables $x_1, \ldots, x_{n-r}$, where $r = \mathrm{Krull.dim}(R/I)$, if the injective ring extension $K[x_{n-r+1}, \ldots, x_n] \to R/I$ is integral.

Lemma 2.1.5 ([6, Lemma 4.1]; see also [40, Proposition 5.4])**.** *Let I be a homogeneous ideal of R, and let $\prec$ be the graded reverse lexicographical ordering on R with $x_n \prec \cdots \prec x_1$. Then, the following are equivalent:*

(1) I is in Noether position with respect to the variables $x_1, \ldots, x_{n-r}$.

(2) For each i with $1 \leq i \leq n - r$, there exists $r_i \geq 0$ such that $x_i^{r_i} \in \langle \mathrm{LM}_{\prec}(I) \rangle$.

(3) $\text{Krull.dim}(R/(I + \langle x_{n-r+1}, \ldots, x_n \rangle)) = 0$.

(4) $\text{Krull.dim}(R/(\langle \text{LM}_{\prec}(I) \rangle + \langle x_{n-r+1}, \ldots, x_n \rangle)) = 0$.

In the proof of [6, Lemma 4.1], the authors first proved the equivalence of the conditions (2), (3), and (4), and the implication (1) $\Rightarrow$ (2). The non-trivial part is that (2), (3), and (4) imply (1), which follows from [40, Proposition 5.4].

Here, we also recall the definition introduced by Hashemi [30] that a homogeneous ideal is in strong Noether position.

Definition 2.1.6 ([30, Definition 2.4 and Theorem 2.19])**.** A homogeneous ideal I of R is said to be in *strong Noether position* (SNP) with respect to the ordered variables $x_1, \ldots, x_n$ if we have $I^{\text{sat}} = (I : x_n^{\infty})$ and

$$(I + \langle x_{n-i+1}, \ldots, x_n \rangle)^{\text{sat}} = ((I + \langle x_{n-i+1}, \ldots, x_n \rangle) : x_{n-i}^{\infty})$$

for any i with $1 \leq i \leq r - 1$.

For equivalent definitions, see [30, Section 2] for details. Note that I is in SNP if and only if $\langle \text{LM}_{\prec}(I) \rangle$ is in SNP (cf. [30, Proposition 2.11]). Hashemi also proved the following inequality:

Theorem 2.1.7 ([30, Proposition 4.7 and Theorem 4.17])**.** *If I is in SNP, then* $\text{reg}(I)$ *is equal to*

$$\overline{\text{hilb}}(I) := \max\{\text{hilb}(I)\} \cup \{\text{hilb}(\langle I, x_{n-i+1}, \ldots, x_n \rangle) \mid 1 \leq i \leq \text{Krull.dim}(R/I)\},$$

which is called the stabilized Hilbert regularity of I, and moreover

$$\text{max.GB.deg}_{\prec}(I) \leq \overline{\text{hilb}}(I)$$

for the graded reverse lexicographical ordering $\prec$ on R with $x_n \prec \cdots \prec x_1$.

Lemma 2.1.8 ([31, Lemma 23] and [32, Proposition 3.2 (v)])**.** *If $r = 1$, then the following are all equivalent:*

(1) I is in Noether position with respect to $x_1, \ldots, x_{n-1}$ (or one of the equivalent conditions given in Lemma 2.1.5 holds).

(2) I is in quasi stable position with respect to the ordered variables $x_1, \ldots, x_n$ (see e.g., [31, Definition 6] for definition).

(3) I is in strong Noether position.

Proof. The equivalence of (1) and (2) is proved in [31, Lemma 23], and that of (2) and (3) follows from [32, Proposition 3.2 (v)]. ☐

By Theorem 2.1.7 and Lemma 2.1.8, we obtain the following theorem:

Theorem 2.1.9 ([31, Proof of Theorem 30]). *Assume that $r = 1$. If $R/(I + \langle x_n \rangle)$ is Artinian (or one of the equivalent conditions given in Lemma 2.1.8 holds), then we have the inequality (1.2) for the graded reverse lexicographical ordering $\prec$ on R with $x_n \prec \cdots \prec x_1$.*

Here, we recall a fact that, if K contains sufficiently many elements, the assumption $\mathrm{Krull.dim}(R/(I + \langle x_n \rangle)) = 0$ in Theorem 2.1.9 is satisfied after a suitable linear changer of coordinates. In the rest of this paper, we explain this fact in details, since we could not find any appropriate reference.

Although it may be an easy consequence of the theory of hyperplane sections, there exists a hyperplane defined over K that does not pass through any of given finite points in a projective space over $\overline{K}$ if K is large enough. Here, we prove a somehow detailed argument:

Lemma 2.1.10. *Let $\boldsymbol{p}_1, \ldots, \boldsymbol{p}_N$ be mutually distinct points in $\mathbb{P}^{n-1}(\overline{K})$. If the cardinality of K is greater than N, then there exists a linear form $\ell \in R$ that does not vanish at any of $\boldsymbol{p}_1, \ldots, \boldsymbol{p}_N$.*

Proof. For each point $\boldsymbol{p}_k = (p_{1,k} : \cdots : p_{n,k}) \in \mathbb{P}^{n-1}(\overline{K})$, we denote by S_k the set of all linear forms that vanish at $\boldsymbol{p}_k$, say

$$S_k = \{ y_1 x_1 + \cdots + y_n x_n \mid y_i \in K \text{ and } y_1 p_{1,k} + \cdots + y_n p_{n,k} = 0 \},$$

which is isomorphic, as K-linear spaces, to the following K-linear space of K^n:

$$S'_k = \{ (y_1, \ldots, y_n) \in K^n \mid p_{1,k} y_1 + \cdots + p_{n,k} y_n = 0 \}.$$

The dimension of S'_k as a K-linear space is at most $n-1$. Indeed, for an arbitrary non-zero entry $p_{i,k}$ of $\boldsymbol{p}_k$, the vector with 1 in the i-th entry and 0's elsewhere does not belong to S'_k, so that S'_k is a proper subspace of K^n.

When K is an infinite field, it follows from the irreducibility of an affine space over K that $\bigcup_{k=1}^{N} S'_k$ cannot cover K^n. When K is a finite field of order q, we have

$$\# \left(\bigcup_{k=1}^{N} S'_k \right) \leq \sum_{k=1}^{N} \# S'_k \leq N \cdot q^{n-1} < q^n,$$

whence a desired linear form over K exists. $\square$

Lemma 2.1.11. *With notation as above, assume that $\mathrm{Krull.dim}(R/I) \leq 1$. If the cardinality of K is greater than $\# V_{\overline{K}}(I)$, then there exists a linear form $\ell \in R$ such that $\mathrm{Krull.dim}(R/(I + \langle \ell \rangle)) = 0$. Moreover, there also exists a suitable linear coordinate change σ sending ℓ to x_n such that $\mathrm{Krull.dim}(R/(I^\sigma + \langle x_n \rangle)) = 0$.*

Proof. The first part follows from Lemma 2.1.10. There exists a $k \in \{1, \ldots, n\}$ such that the x_k-coefficient of ℓ is not zero. We may also assume that the x_k-coefficient is 1, without loss of generality. Let σ_1 be the linear transformation of variables $x_1, \ldots, x_n$ defined by a permutation exchanging x_k and x_n, and write the image of ℓ by σ_1 as $\ell^{\sigma_1} = a_0 x_0 + a_1 x_1 + \cdots + a_n x_n$ for $a_i \in K$ with $a_n = 1$. Then, let σ_2 be a linear transformation of variables $x_1, \ldots, x_n$ defined by

$$h^{\sigma_2} := h(x_1, \ldots, x_{n-1}, x_n - a_1 x_1 - \cdots - a_{n-1} x_{n-1})$$

for $h \in R$. Then, the invertible transformation $\sigma := \sigma_2 \circ \sigma_1$ of variables $x_1, \ldots, x_n$ sends ℓ to x_n, so that $\mathrm{Krull.dim}(R/(I^\sigma + \langle x_n \rangle)) = 0$. $\square$

Remark 2.1.12. In Lemma 2.1.11, we have an exact sequence

$$(R/I)_{d'-1} \xrightarrow{\times \ell} (R/I)_{d'} \longrightarrow (R/(I + \langle \ell \rangle))_{d'} \longrightarrow 0.$$

for each $d' \geq 1$. Therefore, the minimal d for which the multiplication-by-ℓ map $(R/I)_{d'-1} \to (R/I)_{d'}$ is surjective for any $d' \geq d$ is equal to $d_{\mathrm{reg}}(I + \langle \ell \rangle)$.

2.2 Generalized semi-regular sequences

In this subsection, we recall the definition of cryptographic semi-regular sequences and that of generalized ones. Throughout this subsection, let $f_1, \ldots, f_m \in R$ be *homogeneous* polynomials of *positive* degrees $d_1, \ldots, d_m$, and put $I = \langle f_1, \ldots, f_m \rangle$.

Theorem 2.2.1 (cf. [15, Theorem 1])**.** *For each d with $d \geq \max\{d_i : 1 \leq i \leq m\}$, the following are equivalent:*

(1) For each i with $1 \leq i \leq m$, if a homogeneous polynomial $g \in R$ satisfies $g f_i \in \langle f_1, \ldots, f_{i-1} \rangle_R$ and $\deg(g f_i) < d$, then we have $g \in \langle f_1, \ldots, f_{i-1} \rangle_R$, namely, the multiplication-by-f_i map $(R/\langle f_1, \ldots, f_{i-1} \rangle)_{t-d_i} \longrightarrow (R/\langle f_1, \ldots, f_{i-1} \rangle)_t$ is injective for any t with $d_i \leq t < d$.

(2) We have

$$\mathrm{HS}_{R/I}(z) \equiv \frac{\prod_{j=1}^{m} (1 - z^{d_j})}{(1 - z)^n} \pmod{z^d}, \tag{2.2.1}$$

where HS *denotes the Hilbert series.*

(3) We have $H_1(K_\bullet(f_1, \ldots, f_m))_{\leq d-1} = 0$, where $K_\bullet(f_1, \ldots, f_m)$ denotes the Koszul complex on the sequence $(f_1, \ldots, f_m)$.

Definition 2.2.2 ([2, Definition 3]; see also [15, Definition 1])**.** For each d with $d \geq \max\{d_i : 1 \leq i \leq m\}$, a sequence $(f_1, \ldots, f_m)$ of homogeneous polynomials in $R \smallsetminus K$ is said to be d-regular if the equivalent conditions in Theorem 2.2.1 hold.

Here, we define a cryptographic semi-regular sequence:

Definition 2.2.3 ([2, Definition 5], [4, Definition 5]; see also [15, Section 2])**.** A sequence $(f_1, \ldots, f_m)$ of homogeneous polynomials in $R \smallsetminus K$ is said to be *cryptographic semi-regular* if it is $d_{\mathrm{reg}}(I)$-regular.

The Hilbert series of R/I for a cryptographic semi-regular sequence $(f_1, \ldots, f_m)$ is easily computed, by the following Proposition 2.2.4:

Proposition 2.2.4 ([15, Proposition 1 (d)]; see also [4, Proposition 6])**.** *With the same notation as in Theorem 2.2.1, a sequence $(f_1, \ldots, f_m)$ of homogeneous polynomials in $R \smallsetminus K$ is cryptographic semi-regular if and only if*

$$\mathrm{HS}_{R/\langle f_1,\ldots,f_m\rangle}(z) = \left[\frac{\prod_{j=1}^{m}(1-z^{d_j})}{(1-z)^n}\right], \tag{2.2.2}$$

where $[\cdot]$ means truncating a formal power series over $\mathbb{Z}$ after the last consecutive positive coefficient. (Note that, in this case, we have $d_{\mathrm{reg}}(I) = \deg(\mathrm{HS}_{R/I}(z)) + 1$.)

The above 'standard' form (2.2.2) of Hilbert series was introduced by Fröberg in his celebrated paper [23]. We will recall his famous conjecture (Conjecture 2.3.7 below) in the next subsection.

Remark 2.2.5. In [25, Definition 1.3], Fröberg-Hollman already defined a *sufficiently generic ideal* as one generated by homogeneous polynomials $f_1, \ldots, f_m \in R$ satisfying (2.2.2). Namely, a cryptographic semi-regular sequence is nothing but a sequence of homogeneous polynomials generating a sufficiently generic ideal.

Here, we recall the following lemma by Fröberg:

Lemma 2.2.6 ([23, b) on page 121])**.** *With notation as above, we have*

$$\mathrm{HS}_{R/I}(z) \geq^{\mathrm{lex}} \left[\frac{\prod_{j=1}^{m}(1-z^{d_j})}{(1-z)^n}\right]$$

in the lexicographic sense, i.e., the first coefficient that differs, if any, is greater on the left hand side, see [23, page 118].

Remark 2.2.7. For any non-negative integer d, it follows from Lemma 2.2.6 that

$$\left(\mathrm{HS}_{R/I}(z) \bmod z^d\right) \geq^{\mathrm{lex}} \left(\frac{\prod_{j=1}^{m}(1-z^{d_j})}{(1-z)^n} \bmod z^d\right).$$

To deal with the case where R/I has Krull dimension one, we extend the notion of cryptographic semi-regular as follows:

Definition 2.2.8 ([36, Definition 2.3.3])**.** A sequence $(f_1,\ldots,f_m)$ of homogeneous polynomials in $R \smallsetminus K$ is said to be *generalized cryptographic semi-regular* if it is $\widetilde{d}_{\mathrm{reg}}(I)$-regular.

Remark 2.2.9 (cf. [36, Section 4.1])**.** A generalized cryptographic semi-regular (but neither regular nor semi-regular) sequence is easily generated. For example, choosing coefficients of $g_1,\ldots,g_m$ with $m \geq n$ randomly at random from a finite field $K = \mathbb{F}_q$ and choosing an arbitrary point $\boldsymbol{p} = (p_1,\ldots,p_{n-1},p_n) \in \mathbb{A}^n(K)$ with $p_n = 1$, then it is expected that $(f_1,\ldots,f_m)$ with $f_j := g_j(x_1,\ldots,x_n) - g_j(p_1,\ldots,p_n)x_n^{d_j}$ is generalized cryptographic semi-regular (but neither regular nor semi-regular). Without loss of generality, we may take $\boldsymbol{o} = (0:\cdots:0:1)$ as $\boldsymbol{p}$, see Section 4 below for details.

Example 2.2.10. Let $K = \mathbb{F}_{31}$, $n = 3$, and $R = K[x_1,x_2,x_3]$. Here we consider the case where $(d_1,d_2,d_3) = (3,2,2)$ with $m = 3$, and let

$$\begin{aligned}
f_1 &= 27x_1^3 + 5x_1^2x_2 + 6x_1x_2^2 + 20x_2^3 + 29x_1x_2x_3 + 25x_2^2x_3 + 3x_1x_3^2 + 26x_2x_3^2,\\
f_2 &= 2x_1^3 + 29x_1^2x_2 + 25x_1x_2^2 + 3x_2^3 + 25x_1^2x_3 + 20x_1x_2x_3 + 25x_2^2x_3 + 19x_1x_3^2 + 24x_3x_3^2,\\
f_3 &= 12x_1^2 + 12x_1x_2 + 23x_2^2 + x_1x_3 + 10x_2x_3.
\end{aligned}$$

Putting $I = \langle f_1,f_2,f_3\rangle$, one can verify that

$$\mathrm{HS}_{R/I}(z) = 1 + 3z + 5z^2 + 5z^3 + 3z^4 + z^5 + z^6 + \cdots$$

with $\mathrm{Krull.dim}(R/I) = 1$, $D := \mathrm{hilb}(I) = 5$, and $\dim_K(R/I)_D = 1$, so that (f_1,f_2,f_3) is D-regular (in fact $D+1$-regular). Namely (f_1,f_2,f_3) is generalized cryptographic semi-regular, but is neither regular nor cryptographic semi-regular.

Lemma 2.2.11. *If the sequence $\boldsymbol{F} = (f_1,\ldots,f_m)$ is generalized cryptographic semi-regular, then we have*

$$\widetilde{d}_{\mathrm{reg}}(I) \leq \deg\left(\left[\frac{\prod_{j=1}^m (1-z^{d_j})}{(1-z)^n}\right]\right) + 1. \tag{2.2.3}$$

Proof. By our assumption that $\boldsymbol{F}$ is generalized cryptographic semi-regular, the equality (2.2.1) holds for $d = \widetilde{d}_{\mathrm{reg}}(I)$. For any d' with $d' < \widetilde{d}_{\mathrm{reg}}(I)$, we have $\mathrm{HF}_{R/I}(d') > 0$, whence the $z^{d'}$-coefficient of $(\prod_{j=1}^m(1-z^{d_j}))/(1-z)^n$ is also positive. Thus, the inequality (2.2.3) holds. $\square$

For a sequence $\boldsymbol{F} = (f_1,\ldots,f_m)$ of homogeneous polynomials in $R \smallsetminus K$, we consider the following four conditions:

(A) $\boldsymbol{F}$ is regular, equivalently $\mathrm{HS}_{R/I}(z) = (\prod_{j=1}^m(1-z^{d_j}))/(1-z)^n$.

(B) $\boldsymbol{F}$ is cryptographic semi-regular (i.e., $d_{\mathrm{reg}}(I)$-regular).

(C) $\boldsymbol{F}$ is generalized cryptographic semi-regular (i.e., $\widetilde{d}_{\mathrm{reg}}(I)$-regular).

(D) $\boldsymbol{F}$ is $\mathrm{hilb}(I)$-regular.

Lemma 2.2.12. *For conditions (A), (B), (C), and (D) as defined above, there is an implication:*

$$(A) \Rightarrow (B) \Rightarrow (C) \Rightarrow (D), \text{ but } (D) \overset{r \geq 2}{\not\Rightarrow} (C) \overset{r=1}{\not\Rightarrow} (B) \overset{m>n}{\not\Rightarrow} (A).$$

Namely, (A) and (B) are equivalent for $m \leq n$, (B) and (C) are equivalent for $r \neq 1$, (C) and (D) are equivalent for $r \leq 1$, and (A), (B), (C), and (D) are all equivalent for $m = n$ and $r = 0$.

Proof. It is clear that $(A) \Rightarrow (B) \Rightarrow (C) \Rightarrow (D)$. When $r \leq 1$, it follows from (2.1.1) that $(D) \Rightarrow (C)$ holds. Also when $r \neq 1$, we have $d_{\mathrm{reg}}(I) = \widetilde{d}_{\mathrm{reg}}(I)$, whence $(C) \Rightarrow (B)$ holds. For $m \leq n$, Proposition 2.2.4 implies $(B) \Rightarrow (C)$.

On the other hand, it is clear that (B) does not imply (A) if $m > n$. The sequence (f_1, f_2, f_3) provided in Example 2.2.10 gives a counterexample for the implication (C) $\Rightarrow$ (B) when $r = 1$. As for (D) $\Rightarrow$ (C), here is a counterexample with $r = 2$: Let $(g_1, g_2) := (x^3, xy) \in \mathbb{Q}[x, y, z]^2$ and put $I := \langle g_1, g_2 \rangle$. Then one can verify that $\mathrm{HS}_{R/I}(z) = 1 + 3z + 5z^2 + 6z^3 + 7z^4 + \cdots$ with $\mathrm{Krull.dim}(R/I) = 2$ and $D := \mathrm{hilb}(I) = 2$, whence (g_1, g_2) is D-regular but not regular. $\square$

2.3 Genericness and Fröberg's conjecture

Given m positive integers $d_1, \ldots, d_m$, let $V_{n,m,(d_1,\ldots,d_m)} := R_{d_1} \times \cdots \times R_{d_m}$, which is topologically identified with the affine space $\mathbb{A}^{N_1}(K) \times \cdots \times \mathbb{A}^{N_m}(K) = \mathbb{A}^N(K)$ of dimension $N := \sum_{j=1}^{m} N_j$ with $N_j := \binom{n+d_j-1}{d_j}$. Here, we identify a form in R_{d_j} with a point in $\mathbb{A}^{N_j}(K)$ by taking the coefficients. We may write K^N simply for $\mathbb{A}^N(K)$. We also correspond a point $\boldsymbol{F} = (f_1, \ldots, f_m) \in V_{n,m,(d_1,\ldots,d_m)}$ to an ideal $\langle \boldsymbol{F} \rangle_R := \langle f_1, \ldots, f_m \rangle_R$, where the subscript R may be omitted if the meaning is clear from the context.

Following Fröberg's definition (cf. [23], [24], [27]), we say that a form of R_d is *generic* if all monomials of degree d occur with nonzero coefficient and if all the coefficients are algebraically independent over the prime field of K. We call $\boldsymbol{F} = (f_1, \ldots, f_m) \in V_{n,m,(d_1,\ldots,d_m)}$ a *generic* sequence if all f_i's are generic forms with coefficients that are mutually algebraic independent over the prime field. If K does not contain N elements that are algebraically independent over the prime field of K (for example K is an algebraic extension of the prime field), we may replace K by its arbitrary extension field L containing such N elements. When we consider a property of Hilbert series, this makes a sense since every Hilbert series is not changed by extending the field of coefficients.

On the other hand, the genericness of a property for $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ is defined as follows:

Definition 2.3.1 (cf. [43, Section 1])**.** For given n, m, and $(d_1, \ldots, d_m)$, a property $\mathcal{P}$ for $\boldsymbol{F} = (f_1, \ldots, f_m) \in V_{n,m,(d_1,\ldots,d_m)}$ is said to be *generic* if it holds on a non-empty Zariski open set in $V_{n,m,(d_1,\ldots,d_m)}$, namely there exists a non-empty Zariski open set U in $V_{n,m,(d_1,\ldots,d_m)}$ such that

$$U \subset \{\boldsymbol{F} = (f_1, \ldots, f_m) \in V_{n,m,(d_1,\ldots,d_m)} : \boldsymbol{F} \text{ satisfies } \mathcal{P}\}.$$

Here, we recall Fröberg's important lemma for the minimality of the Hilbert series on generic sequences. Note that the characteristic of K is assumed to be 0 in the original paper [23], but it can be easily proved that the assertion of the lemma holds for any characteristic. We here state the lemma in a slightly modified (but equivalent) form for the convenience of our use.

Lemma 2.3.2 ([23, Lemma 1])**.** *With notation as above, let Ω be an extension field of K, and let $\mathcal{C} := \{c_{j,t} : 1 \leq j \leq m,\ t \in \mathcal{T}_{d_j}\}$ be a set of elements in Ω that are algebraically independent over the prime field K_0 of K. Here $\mathcal{T}_d$ is the set of monomials of degree d in $\{x_1, \ldots, x_n\}$. Let $L := K(\mathcal{C})$, let $R_L := R \otimes_K L = L[x_1, \ldots, x_n]$, and put $g_j := \sum_{t \in \mathcal{T}_{d_j}} c_{j,t} t \in (R_L)_{d_j}$ for each $1 \leq j \leq m$. Namely $\boldsymbol{G} = (g_1, \ldots, g_m)$ is a* generic sequence*. Then, for any $\boldsymbol{F} = (f_1, \ldots, f_m) \in V_{n,m,(d_1,\ldots,d_m)}$, we have the coefficient-wise inequality*

$$\mathrm{HS}_{R/\langle f_1,\ldots,f_m\rangle_R}(z) = \mathrm{HS}_{R_L/\langle f_1,\ldots,f_m\rangle_{R_L}} \geq \mathrm{HS}_{R_L/\langle g_1,\ldots,g_m\rangle_{R_L}}(z). \tag{2.3.1}$$

Remark 2.3.3. Since every Hilbert series is not changed by extending the field of coefficients, the Hilbert series of the quotient by an ideal generated by a generic sequence depends only on $\mathrm{char}(K)$, n, m, and $(d_1, \ldots, d_m)$, see e.g., [16, Lemma and Definition 4]. Thus, in discussing the Hilbert series, we may consider the ideal generated by $\boldsymbol{G}$ over $K_0(\mathcal{C})[X]$ instead of that over $K(\mathcal{C})[X]$. Because, the reduced Gröbner basis of $\langle \boldsymbol{G} \rangle_{R_L}$ coincides with that of $\langle \boldsymbol{G} \rangle_{K_0(\mathcal{C})[X]}$, since the well-known Buchberger's algorithm can be performed over a field, over which the generating set is defined. Hence the minimal generating set of $\mathrm{in}(\langle \boldsymbol{G} \rangle_{R_L})$ coincides with that of $\mathrm{in}(\langle \boldsymbol{G} \rangle_{K_0(\mathcal{C})[X]})$. This also implies that, for computing the reduced Gröbner basis of $\langle \boldsymbol{G} \rangle_{R_L}$, we may assume that all elements of $\mathcal{C}$ are algebraically independent over K and compute it over $L = K(\mathcal{C})$.

If K is an infinite field, Fröberg-Löfwall proved in [26, Theorem 1] that there exists a non-empty Zariski-open subset $U_{\min} \subset V_{n,m,(d_1,\ldots,d_m)}$ on which the quotients by the corresponding ideals have the same Hilbert series, which is minimal in the coefficient-wise sense. Note that, even if K contains the set $\mathcal{C}$ of N algebraically independent elements, $\boldsymbol{F} \in U_{\min}$ not necessarily implies the genericness of $\boldsymbol{F}$.

On the other hand, the Hilbert series of $R/\langle \boldsymbol{F} \rangle$ for $\boldsymbol{F} \in U_{\min}$ is equal to that of $R_L/\langle \boldsymbol{G} \rangle$ for a generic sequence $\boldsymbol{G}$ over L as in Lemma 2.3.2. The existence of the non-empty Zariski open set $U_{\min}$ can be easily verified by considering the *stability* of the reduced Gröbner basis in the context of so-called a *comprehensive Gröbner system for a parametric ideal.* A result by Nabeshima in [42] on "generic Gröbner basis of a parametric ideal" can be translated directly to our problem where the generic sequence $\boldsymbol{G}$ is given from $K[\mathcal{C}][x_1, \dots, x_n]$ and $\mathcal{C}$ is the set of *coefficient parameters.* Here we assume that all elements of $\mathcal{C}$ are algebraically independent over K and hence they are treated as independent indeterminates. (See Remark 2.3.3.) In the below, we show a useful result for our purpose. We let $\prec_1$ be a block ordering with $X := \{x_1, \dots, x_n\} \succ\succ \mathcal{C}$ such that its restriction on X coincides with the given ordering $\prec$. Let $\boldsymbol{H}$ be a finite set of $K[\mathcal{C}][X]$ and $\mathrm{GB}(\boldsymbol{H})$ the reduced Gröbner basis of the ideal $\langle \boldsymbol{H} \rangle_{L[x]}$ with respect to the restriction $\prec$ of $\prec_1$ on the monomials in X, where $L = K(\mathcal{C})$. Moreover, let $\widetilde{\mathrm{GB}}(\boldsymbol{H}) := \{\mathrm{dlcm}(h)h : h \in \mathrm{GB}(\boldsymbol{H})\}$ and $\ell(\boldsymbol{H}) = \prod_{h' \in \widetilde{\mathrm{GB}}(\boldsymbol{H})} \mathrm{LC}_{\prec}(h')$ in $K[\mathcal{C}][X]$, where $\mathrm{dlcm}(h)$ denotes the least common multiple of all denominators of coefficients in $K(\mathcal{C})$ of h and each $h' \in \widetilde{\mathrm{GB}}(\boldsymbol{H})$ is considered as an element in $K[\mathcal{C}][X]$, that is, a polynomial in X over $K[\mathcal{C}]$. For $\boldsymbol{a} \in \mathbb{A}^N(K)$, we denote by $\pi_{\boldsymbol{a}}$ the substitution homomorphism from $K[\mathcal{C}][X]$ to $R = K[X]$, that is, $\pi_{\boldsymbol{a}}(h(\mathcal{C}, X)) = h(\boldsymbol{a}, X)$ for $h \in K[\mathcal{C}, X]$. For $h \in L[X]$, when the denominator of each coefficient of h does not vanish at $\boldsymbol{a}$, the image $\pi_{\boldsymbol{a}}(h)$ is defined naturally.

Theorem 2.3.4 ([42, Theorem 3])**.** *With notation as above, consider $\boldsymbol{H}$ and $\widetilde{\mathrm{GB}}(\boldsymbol{H})$ to be subsets of $K[\mathcal{C} \cup X]$, and let $S(\boldsymbol{H})$ be the reduced Gröbner basis of the ideal quotient $\langle \boldsymbol{H} \rangle_{K[\mathcal{C} \cup X]} : \langle \widetilde{\mathrm{GB}}(\boldsymbol{H}) \rangle_{K[\mathcal{C} \cup X]}$ with respect to $\prec_1$. Then, we have $S(\boldsymbol{H}) \cap K[\mathcal{C}] \neq \emptyset$, and for any $\boldsymbol{a}$ in $\mathbb{A}^N(\overline{K}) \setminus (V_{\overline{K}}(S(\boldsymbol{H}) \cap K[\mathcal{C}]) \cup V_{\overline{K}}(\ell(\boldsymbol{H})))$, the set $\pi_{\boldsymbol{a}}(\mathrm{GB}(\boldsymbol{H}))$ is the reduced Gröbner basis of $\langle \pi_{\boldsymbol{a}}(\boldsymbol{H}) \rangle_{K[X]}$ with respect to $\prec$.*

Now we return to our case where $\boldsymbol{H} = \boldsymbol{G}$. Let $\mathcal{O} = \mathbb{A}^N(K) \setminus (V_K(S(\boldsymbol{G}) \cap K[\mathcal{C}]) \cup V_K(\ell(\boldsymbol{G})))$. Then $\mathcal{O}$ is Zariski-open in $\mathbb{A}^N(K)$. Moreover, $\mathcal{O}$ is not empty, since $\langle S(\boldsymbol{G}) \cap K[C] \rangle \cap \langle \ell(\boldsymbol{G}) \rangle$ is a non-zero ideal and since K is an infinite field. It is obvious that for any $\boldsymbol{a} \in \mathcal{O}$, $\pi_{\boldsymbol{a}}(\mathrm{GB}(\boldsymbol{G}))$ is the reduced Gröbner basis of $\langle \pi_{\boldsymbol{a}}(\boldsymbol{G}) \rangle_R$ with respect to $\prec$. Moreover, for such $\boldsymbol{a}$, since $\mathrm{LC}_{\prec}(\pi_{\boldsymbol{a}}(h)) = 1$ for any $h \in \mathrm{GB}(\boldsymbol{G})$, we have $\mathrm{LM}_{\prec}(\pi_{\boldsymbol{a}}(\mathrm{GB}(\boldsymbol{G}))) = \mathrm{LM}_{\prec}(\mathrm{GB}(\boldsymbol{G}))$.

Corollary 2.3.5. *Under the assumption in Theorem 2.3.4, there is a non-empty Zariski open set $\mathcal{O}$ in K^N such that, for any $\boldsymbol{a}$ in $\mathcal{O}$, the set of leading monomials of the reduced Gröbner basis of $\langle \pi_{\boldsymbol{a}}(\boldsymbol{G}) \rangle_R$ coincides with that of $\langle \boldsymbol{G} \rangle_{R_L}$.*

For each ideal I of $K[X]$, the Hilbert series $\mathrm{HS}_{R/I}(z)$ is determined by its initial ideal and so by the leading monomials of its (reduced) Gröbner basis. Thus, Corollary 2.3.5 implies that $\mathrm{HS}_{R/\langle \pi_{\boldsymbol{a}}(\boldsymbol{G}) \rangle}(z)$ coincides with $\mathrm{HS}_{R_L/\langle \boldsymbol{G} \rangle}(z)$ for any $\boldsymbol{a} \in \mathcal{O}$. Considering the corresponding set U of $\mathcal{O}$ in $V_{n,m,(d_1,\dots,d_m)}$, we have the following proposition and a conjecture. Here, we remark that the following proposition may be a well-known fact. However, the authors could not find any reference for its proof, and therefore they would write a proof here for the readers' convenience; the idea of the proof will be used to obtain an analogous result in Section 4 below.

Proposition 2.3.6. *Assume that K is an infinite field. Then, for given n, m, and $(d_1, \ldots, d_m)$, the following conditions (1), (2), (3), and (4) are equivalent to each other:*

(1) There exists an $\boldsymbol{F} = (f_1, \ldots, f_m) \in V_{n,m,(d_1,\ldots,d_m)}$ satisfying (2.2.2), i.e., $\boldsymbol{F}$ is cryptographic semi-regular (in other words, the ideal $\langle \boldsymbol{F} \rangle$ is sufficiently generic).

(2) For the non-empty Zariski open set $U_{\min}$ defined in the paragraph just after Remark 2.3.3, any $\boldsymbol{F} = (f_1, \ldots, f_m) \in U_{\min}$ satisfies (2.2.2).

(3) The property that $\boldsymbol{F} = (f_1, \ldots, f_m) \in V_{n,m,(d_1,\ldots,d_m)}$ satisfies (2.2.2) is generic (in terms of Definition 2.3.1).

(4) For $\mathcal{C}$, L, R_L, and $\boldsymbol{G} = (g_1, \ldots, g_m)$ as in Lemma 2.3.2, we have

$$\mathrm{HS}_{R_L/\langle g_1,\ldots,g_m \rangle}(z) = \left[\frac{\prod_{j=1}^{m}(1 - z^{d_j})}{(1 - z)^n} \right]. \tag{2.3.2}$$

Namely, the generic sequence $\boldsymbol{G}$ is cryptographic semi-regular (in other words, the ideal $\langle \boldsymbol{G} \rangle_{R_L}$ is sufficiently generic).

Proof. The implication (2) $\Rightarrow$ (1) is obvious.

To see (1) $\Rightarrow$ (4), it follows from Lemma 2.2.6 and Lemma 2.3.2 that

$$\mathrm{HS}_{R/\langle f_1,\ldots,f_m \rangle}(z) \geq \mathrm{HS}_{R_L/\langle g_1,\ldots,g_m \rangle}(z) \geq^{\mathrm{lex}} \left[\frac{\prod_{j=1}^{m}(1 - z^{d_j})}{(1 - z)^n} \right]. \tag{2.3.3}$$

By our assumption that $\mathrm{HS}_{R/\langle f_1,\ldots,f_m \rangle}(z)$ satisfies (2.2.2), we have (2.3.2).

The implication (4) $\Rightarrow$ (3) follows from Corollary 2.3.5.

The implication (3) $\Rightarrow$ (2) follows from the irreducibility of $V_{n,m,(d_1,\ldots,d_m)}$. □

Here, we recall *Fröberg's conjecture*:

Conjecture 2.3.7 (Fröberg's conjecture [23])**.** Assume that K is an infinite field. For any n, m, and $(d_1, \ldots, d_m)$, the equivalent conditions (1), (2), (3), and (4) in Proposition 2.3.6 hold.

See Pardue's paper [43] for some conjectures equivalent to Fröberg's one (Conjecture 2.3.7). He also proved that Moreno-Socías conjecture [41] is stronger than Conjecture 2.3.7, see [43, Theorem 2] for a proof.

3 Proofs of Theorems 1.1 and 1.2

As in the previous sections, let $R = K[x_1, \ldots, x_n]$ be the polynomial ring of n variables over a field K (in any characteristic). Let I be a proper homogeneous ideal of R, and let $F = \{f_1, \ldots, f_m\}$ be a homogeneous subset of R generating I. For each $1 \leq j \leq m$, put $d_j := \deg(f_j)$ (total degree).

3.1 Proof of Theorem 1.1

We first prove Theorem 1.1 (1), which can be easily proved by Lemma 2.2.11 together with the following easy lemma in the case where $m = n - 1$:

Lemma 3.1.1. *With notation as above, if $m = n - 1$ and if the sequence $(f_1, \ldots, f_m, x_n)$ is cryptographic semi-regular (equivalently, it is regular since $m + 1 = n$), then we have $\widetilde{d}_{\mathrm{reg}}(I) = d_{\mathrm{reg}}(I + \langle x_n \rangle_R) - 1$, and hence*

$$\mathrm{max.GB.deg}_{\prec}(I) \leq \max\{d_{\mathrm{reg}}(I+\langle x_n\rangle), \widetilde{d}_{\mathrm{reg}}(I)\} = d_{\mathrm{reg}}(I+\langle x_n\rangle) = \sum_{j=1}^{n-1}(d_j - 1) + 1$$

for the graded reverse lexicographical ordering $\prec$ with $x_n \prec \cdots \prec x_1$. Moreover, $(f_1, \ldots, f_m)$ is generalized cryptographic semi-regular (equivalently, it is regular since $m < n$).

Proof. Since $(f_1, \ldots, f_m, x_n)$ is regular, $(f_1, \ldots, f_m)$ is also regular, so that

$$\mathrm{HS}_{R/I}(z) = \mathrm{HS}_{R/(I+\langle x_n\rangle)}(z) \cdot \frac{1}{1-z} = \mathrm{HS}_{R/(I+\langle x_n\rangle)}(z) \cdot (1 + z + z^2 + \cdots)$$

with

$$\mathrm{HS}_{R/(I+\langle x_n\rangle)}(z) = \frac{\prod_{j=1}^{n-1}(1 - z^{d_j}) \cdot (1-z)}{(1-z)^n} = \frac{\prod_{j=1}^{n-1}(1 - z^{d_j})}{(1-z)^{n-1}} = \prod_{j=1}^{n-1}\left(\sum_{k=0}^{d_j - 1} z^k\right),$$

which is a polynomial. It is clear that $\widetilde{d}_{\mathrm{reg}}(I) = d_{\mathrm{reg}}(I + \langle x_n \rangle_R) - 1$, as desired. $\square$

Proof of Theorem 1.1 (1): If $m \geq n$, our assumption that $(f_1, \ldots, f_m, x_n)$ is cryptographic semi-regular implies

$$d_{\mathrm{reg}}(I + \langle x_n \rangle) = \deg\left(\left[\frac{\prod_{j=1}^{m}(1 - z^{d_j})}{(1-z)^{n-1}}\right]\right) + 1,$$

whose right hand side is upper-bounded by $D^{(n,m)}$ for $m \geq n$. On the other hand, since $(f_1, \ldots, f_m)$ is generalized cryptographic semi-regular, it follows from

Lemma 2.2.11 that we have the inequality (2.2.3), the right hand side of which is nothing but $D^{(n,m)}$ for $m \geq n$. Therefore, we obtain (1.3).

If $m = n-1$, it follows from Lemma 3.1.1 that $\max\{d_{\mathrm{reg}}(I + \langle x_n \rangle), \widetilde{d}_{\mathrm{reg}}(I)\}$ is upper-bounded by $\sum_{j=1}^{n-1}(d_j - 1) + 1$, which is equal to $D^{(n,m)}$ for $m = n-1$. $\square$

Next, we shall prove (2) of Theorem 1.1. For this, we estimate the complexity of computing a Gröbner basis of a homogeneous ideal with Macaulay matrices.

Let $\prec$ be an *arbitrary* graded monomial ordering on R. It is well-known that the complexity of the Gröbner basis computation for the homogeneous set F is estimated as that of computing the reduced row echelon form (RREF) of the degree-D *Macaulay matrix*, denoted by $M_{\leq D}(F)$, where D is an arbitrary upper-bound on $\mathrm{max.GB.deg}_{\prec}(F)$. Here, the degree-$D$ Macaulay matrix $M_{\leq D}(F)$ (in homogeneous case) is a block matrix of the form

$$M_{\leq D}(F) = \begin{pmatrix} M_D(F) & & \\ & \ddots & \\ & & M_{d_0}(F) \end{pmatrix}$$

with $d_0 := \min\{\deg(f_j) : 1 \leq j \leq m\}$, where each $M_d(F)$ is a *homogeneous* Macaulay matrix defined as follows: For each d with $d \geq d_0$, the degree-d homogeneous Macaulay matrix $M_d(F)$ of F has columns indexed by the monomials of R_d sorted, from left to right, according to the chosen order $\prec$. The rows of $M_d(F)$ are indexed by the polynomials $m_{i,j} f_j$, where $m_{i,j} \in R$ is a monomial such that $\deg(m_{i,j} f_j) = d$. For a monomial $t \in \mathcal{T}_d$ of degree d, the $(m_{i,j} f_j, t)$-entry of $M_d(F)$ is defined as the coefficient of t in $m_{i,j} f_j$.

Theorem 3.1.2. *With notation as above, we have the following:*

(1) [3, Proposition 1] The RREF of $M_{\leq D}(F)$ is computed in

$$O\left(mD\binom{n+D-1}{D}^{\omega}\right). \tag{3.1.1}$$

Hence, if $\prec$ is a graded ordering, then the complexity of the Gröbner basis computation for F is upper-bounded by (3.1.1) for $D = \mathrm{max.GB.deg}_{\prec}(F)$.

(2) Let D be a non-negative integer such that $D = O(n)$. Then, the RREF of $M_{\leq D}(F)$ is computed in

$$O\left(m\binom{n+D-1}{D}^{\omega}\right). \tag{3.1.2}$$

Hence, if $\prec$ is a graded ordering, and if $\mathrm{max.GB.deg}_{\prec}(F) = D = O(n)$, then the complexity of the Gröbner basis computation for F is upper-bounded by (3.1.2).

Proof. It suffices to estimate the complexity of computing the RREFs of the homogeneous Macaulay matrices $M_D(F), \ldots, M_{d_0}(F)$. For each d with $d_0 \leq d \leq D$, the number of rows (resp. columns) in $M_d(F)$ is $\displaystyle\sum_{1 \leq i \leq m,\ d \geq d_i} \binom{n-1+d-d_i}{d-d_i}$ (resp. $\binom{n-1+d}{d}$). It follows from $d - d_i \leq d-1$ that

$$\sum_{1 \leq i \leq m,\ d \geq d_i} \binom{n-1+d-d_i}{d-d_i} \leq \sum_{i=1}^{m} \binom{n-1+d-1}{d-1} = m\binom{n-1+d-1}{d-1},$$

whence the number of rows in $M_d(F)$ is upper-bounded by $m\binom{n-1+d-1}{d-1}$. Therefore, the complexity of the row reductions on $M_D(F), \ldots, M_{d_0}(F)$ is upper-bounded by

$$\begin{aligned}\sum_{d=d_0}^{D} m\binom{n-1+d-1}{d-1}\binom{n-1+d}{d}^{\omega-1} &\leq m\sum_{d=1}^{D}\binom{n-1+d-1}{d-1}\binom{n-1+d}{d}^{\omega-1} \\ &\leq m\sum_{d=1}^{D}\binom{n-1+d}{d}^{\omega}, \qquad (3.1.3)\end{aligned}$$

where we used a fact that the reduced row echelon form of a $k \times \ell$ matrix A over a field K can be computed in $O(k\ell^{\omega-1})$, see Remark 3.1.3 below for its proof. Clearly (3.1.3) is upper-bounded by $mD\binom{n+D-1}{D}^{\omega}$, we obtain the estimation (3.1.1) in (1).

In the following, we shall improve the estimation (3.1.1) in (1) to obtain (3.1.2) in (2), assuming that $D = O(n)$. Here, for each d with $1 \leq d \leq D-1$, we have

$$\binom{n-1+d}{d} = \frac{(n-1+d)!}{(n-1)!d!} = \binom{n-1+d+1}{d+1} \times \frac{d+1}{n+d}.$$

Repeating this procedure, we obtain

$$\begin{aligned}&\binom{n-1+d+1}{d+1} \times \frac{d+1}{n+d} = \binom{n-1+d+2}{d+2} \times \frac{d+2}{n+d+1} \times \frac{d+1}{n+d} \\ &= \cdots = \binom{n-1+D}{D} \times \frac{D}{n+D-1} \times \cdots \times \frac{d+2}{n+d+1} \times \frac{d+1}{n+d}.\end{aligned}$$

It is clear that $\frac{d'+1}{n+d'} = 1 - \frac{n-1}{n+d'}$ is monotonically increasing with respect to d'. Putting $c := (D-1)/n$, we have $D = cn+1$, so that

$$\frac{d'+1}{n+d'} \leq \frac{D}{n+D-1} = \frac{cn+1}{(c+1)n} = \frac{c+\frac{1}{n}}{c+1}$$

for any d' with $d' \leq D-1$. As we consider $n \to \infty$, we may suppose that

$$\frac{d'+1}{n+d'} \leq \frac{c+\frac{1}{n}}{c+1} < \frac{c+\varepsilon}{c+1} < 1$$

for some constant ε with $0 < \varepsilon < 1$. Therefore, we obtain

$$\binom{n-1+D}{D} \times \frac{D}{n+D-1} \times \cdots \times \frac{d+2}{n+d+1} \times \frac{d+1}{n+d} < \binom{n-1+D}{D} \left(\frac{c+\varepsilon}{c+1} \right)^{D-d},$$

whence

$$\begin{aligned} \sum_{d=1}^{D} \binom{n-1+d}{d}^{\omega} &< \binom{n-1+D}{D}^{\omega} \sum_{k=0}^{D-1} \left(\frac{c+\varepsilon}{c+1} \right)^{k\omega} \\ &= \binom{n-1+D}{D}^{\omega} \times \frac{1-\left(\frac{c+\varepsilon}{c+1}\right)^{D\omega}}{1-\left(\frac{c+\varepsilon}{c+1}\right)^{\omega}} \\ &< \binom{n-1+D}{D}^{\omega} \times \frac{1}{1-\left(\frac{c+\varepsilon}{c+1}\right)^{\omega}}, \end{aligned}$$

where we put $k := D-d$. It follows from $D = O(n)$ that c is upper-bounded by a constant. By $0 < \varepsilon < 1$, it is clear that $\frac{c+\varepsilon}{c+1}$ is also upper-bounded by a positive constant smaller than 1, whence $1/(1-(\frac{c+\varepsilon}{c+1})^{\omega}) = O(1)$ as $n \to \infty$. We have proved the assertion (2). $\square$

Remark 3.1.3. In the proof of Theorem 3.1.2, we used a fact that the reduced row echelon form of a $k \times \ell$ matrix A over a field K can be computed in $O(k\ell^{\omega-1})$, where k can be greater than ℓ. This fact is proved by considering the following procedures to compute the reduced row echelon form of A:

(0) Write $k = \ell q + r$ for $q, r \in \mathbb{Z}$ with $0 \leq r < \ell$.

(1) Choose and remove 2ℓ rows from A, and let A' be a $2\ell \times \ell$ matrix whose rows are the chosen 2ℓ rows.

(2) Compute the reduced row echelon form of A': Inserting ℓ zeros to the end of each row of A', construct a $2\ell \times 2\ell$ square matrix, and compute its reduced row echelon form.

(3) As a result of (2), we obtain more than or equal to ℓ zero row vectors. We add the nonzero (linearly independent) row vectors obtained to A.

(4) Go back to (1).

Repeating the procedures from (1) to (4) at most q times, we obtain the reduced row echelon form B of the initial A. Since the complexity of (2) is $(2\ell)^{\omega}$, the total complexity to obtain B is $q \cdot (2\ell)^{\omega} \leq (k/\ell) \cdot 2^{\omega}\ell^{\omega} = O(k\ell^{\omega-1})$, as desired.

Proof of Theorem 1.1 (2): For the graded reverse lexicographical ordering $\prec$ on R with $x_n \prec \cdots \prec x_1$, it follows from Theorem 1.1 (1) that $D^{(n,m)}$ gives an upper bound on $\mathrm{max.GB.deg}_{\prec}(F)$. Therefore, a Gröbner basis of $\langle F \rangle_R$ is obtained by computing the RREF of $M_{\leq D}(F)$ for $D = D^{(n,m)}$. Here, the value $D^{(n,m)}$ of (1.4) is maximized at $m = n$: In this case, $D^{(n,m)}$ is equal to the Lazard's bound $\sum_{j=1}^{n}(d_j - 1) + 1$, which is $O(n)$ for fixed $d_1, \ldots, d_m$. Therefore, it follows from Theorem 3.1.2 (2) that the RREF of $M_{\leq D}(F)$ for $D = D^{(n,m)}$ is computed in the complexity (1.5), as desired. $\square$

Example 3.1.4. The bound $D^{(n,m)}$ in (1.3) is optimal for a homogeneous ideal I satisfying the assumptions of Theorem 1.1. For example, considering the case where $K = \mathbb{F}_{31}$ and $(n,m) = (3,4)$, and letting $F = \{f_1, f_2, f_3, f_4\}$ be a generating set for I with

$$\begin{aligned} f_1 &= 3x_1^2 + 20x_1x_2 + x_2^2 + 7x_1x_3 + 15x_2x_3, \\ f_2 &= 18x_1^2 + 15x_1x_2 + 14x_2^2 + 24x_1x_3 + 4x_2x_3, \\ f_3 &= 6x_1^2 + 16x_1x_2 + x_2^2 + x_1x_3 + 8x_2x_3, \\ f_4 &= 21x_1^2 + x_1x_2 + 21x_2^2 + 24x_1x_3 + 6x_2x_3, \end{aligned}$$

we have $\mathrm{Krull.dim}(R/I) = 1$ and

$$\mathrm{HS}_{R/(I+\langle x_3 \rangle)}(z) = 1 + 2z, \quad \mathrm{HS}_{R/I}(z) = 1 + 3z + 2z^2 + z^3 + z^4 + z^5 + \cdots$$

with $d_{\mathrm{reg}}(I + \langle x_3 \rangle) = 2$ and $\widetilde{d}_{\mathrm{reg}}(I) = 3$, and therefore $(f_1, f_2, f_3, f_4, x_3)$ (resp. (f_1, f_2, f_3, f_4)) is cryptographic semi-regular (resp. generalized cryptographic semi-regular). One can easily verify $D^{(3,4)} = 3$. The reduced Gröbner basis of I with respect to the graded reverse lexicographical ordering $\prec$ with $x_3 \prec x_2 \prec x_1$ is

$$\{x_2x_3^2, x_1^2 + 27x_2x_3, x_1x_2 + 14x_2x_3, x_2^2 + 30x_2x_3, x_1x_3 + 26x_2x_3\},$$

whence $\mathrm{max.GB.deg}_{\prec}(I) = 3$. Namely, we have

$$3 = \mathrm{max.GB.deg}_{\prec}(I) = \max\{d_{\mathrm{reg}}(I + \langle x_n \rangle), \widetilde{d}_{\mathrm{reg}}(I)\} = \widetilde{d}_{\mathrm{reg}}(I) = D^{(n,m)}.$$

On the other hand, $\max\{d_{\mathrm{reg}}(I + \langle x_n \rangle), \widetilde{d}_{\mathrm{reg}}(I)\}$ in (1.3) can be strictly less than $D^{(n,m)}$, e.g., when $F := \{f_1 := 23x_1^2 + 26x_1x_2 + 13x_2^2, f_2 := 29x_1^2 + x_1x_2 + x_2^2\}$ with $n = m = 2$ and $K = \mathbb{F}_{31}$, a straightforward computation shows that $I = \langle F \rangle$ satisfies the assumptions of Theorem 1.1 with $\mathrm{max.GB.deg}_{\prec}(I) = 2$ and $d_{\mathrm{reg}}(I + \langle x_2 \rangle) = \widetilde{d}_{\mathrm{reg}}(I) = 2 < D^{(2,2)} = 3$. (In this case, we can replace $D^{(n,m)}$ by $D^{(n,m)} - 1$ in (1.5)).

3.2 Proof of Theorem 1.2

As in the previous subsection, let $\prec$ be the graded reverse lexicographical ordering on $R = K[x_1, \ldots, x_n]$ with $x_n \prec \cdots \prec x_1$. We also denote the restriction of $\prec$ on $K[x_1, \ldots, x_{n-1}]$ by the same notation $\prec$. In the following, we shall prove Theorem 1.2, namely the inequality (1.2) becomes an equality if the initial ideal $\mathrm{in}_{\prec}(I) = \langle \mathrm{LM}_{\prec}(I) \rangle_R$ is weakly reverse lexicographic. Here, a weakly reverse lexicographic ideal is a monomial ideal J such that if x^{α} is one of the minimal generators of J then every monomial of the same degree which preceeds x^{α} must belong to J as well (see also e.g., [43, Section 4] for the definition).

As in [30] and [31] (see also [35] and [36]), we consider the elimination ideal

$$I' := (I + \langle x_n \rangle_R) \cap K[x_1, \ldots, x_{n-1}] = \langle f_1|_{x_n=0}, \ldots, f_m|_{x_n=0} \rangle_{K[x_1,\ldots,x_{n-1}]}.$$

Since $R/(I + \langle x_n \rangle)$ and $K[x_1, \ldots, x_{n-1}]/I'$ are isomorphic as graded rings, they have the same Hilbert series, and $d_{\mathrm{reg}}(I') = d_{\mathrm{reg}}(I + \langle x_n \rangle_R)$. Put $D := d_{\mathrm{reg}}(I') = d_{\mathrm{reg}}(I + \langle x_n \rangle_R)$. By comparing Hilbert series, it can be also easily checked that, the sequence $(f_1, \ldots, f_m, x_n) \in R^{m+1}$ is cryptographic semi-regular if and only if $(f_1|_{x_n=0}, \ldots, f_m|_{x_n=0}) \in K[x_1, \ldots, x_{n-1}]^m$ is cryptographic semi-regular, equivalently these sequences are both D-regular.

Let G (resp. G') be the reduced Gröbner basis of I (resp. I') with respect to $\prec$. Note that, since $I + \langle x_n \rangle_R = I'R + \langle x_n \rangle_R$, the reduced Gröbner basis of $I + \langle x_n \rangle_R$ is given by $G' \cup \{x_n\}$, whence $\mathrm{max.GB.deg}_{\prec}(I') = \mathrm{max.GB.deg}_{\prec}(I + \langle x_n \rangle_R)$.

To prove Theorem 1.2, we start with extending some results in [35] to our case.

Lemma 3.2.1 (cf. [35, Corollary 2]). *With notation as above, we assume that $(f_1, \ldots, f_m, x_n) \in R^{m+1}$ is cryptographic semi-regular (i.e., $(f_1|_{x_n=0}, \ldots, f_m|_{x_n=0}) \in K[x_1, \ldots, x_{n-1}]^m$ is cryptographic semi-regular), namely both the sequences are D-regular with $D := d_{\mathrm{reg}}(I') = d_{\mathrm{reg}}(I + \langle x_n \rangle_R)$. Then, we have*

$$(\langle \mathrm{LM}_{\prec}(I) \rangle_R)_d = (\langle \mathrm{LM}_{\prec}(I') \rangle_R)_d \tag{3.2.1}$$

for any d with $d < D$.

Proof. We first prove $(\langle \mathrm{LM}_{\prec}(I) \rangle_R)_d \supset (\langle \mathrm{LM}_{\prec}(I') \rangle_R)_d$ for any non-negative integer d. For this, it suffices to show that $\mathrm{LM}_{\prec}(f) \in \mathrm{LM}_{\prec}(I)$ for any $f \in I'_d \smallsetminus \{0\}$. We may assume that f is homogeneous, and then it is written as $f = g + x_n h$ for some $g \in I_d$ and $h \in R_{d-1}$. Here, g is not divisible by x_n, since otherwise $f \notin K[x_1, \ldots, x_{n-1}]$, a contradiction. Therefore, we have $\mathrm{LM}_{\prec}(f) = \mathrm{LM}_{\prec}(g)$ with $g \in I$, as desired.

Since $(f_1, \ldots, f_m, x_n)$ is D-regular, it follows from [15, Proposition 2 a)] (cf. [35, Corollary 1]), its sub-sequence $(f_1, \ldots, f_m)$ is also D-regular. Hence, we have

$$\begin{aligned}
\mathrm{HS}_{R/I}(z) &\equiv \frac{\prod_{j=1}^m (1-z^{d_j})}{(1-z)^n} \equiv \frac{\prod_{j=1}^m (1-z^{d_j}) \cdot (1-z)}{(1-z)^n} \cdot \frac{1}{1-z} \\
&\equiv \mathrm{HS}_{R/(I+\langle x_n \rangle)}(z) \cdot \frac{1}{1-z} \equiv \mathrm{HS}_{R/(I+\langle x_n \rangle)}(z) \cdot (1 + z + z^2 + \cdots) \\
&\equiv \mathrm{HS}_{K[x_1,\ldots,x_{n-1}]/I'}(z) \cdot (1 + z + z^2 + \cdots) \pmod{z^D}.
\end{aligned}$$

Thus, for any d with $d < D = d_{\mathrm{reg}}(I')$, one has $\mathrm{HF}_{R/I}(d) = \sum_{i=0}^{d} \mathrm{HF}_{K[x_1,\ldots,x_{n-1}]/I'}(i)$, so that

$$\dim_K R_d - \dim_K I_d = \sum_{i=0}^{d} (\dim_K K[x_1, \ldots, x_{n-1}]_i - \dim_K I'_i). \tag{3.2.2}$$

Since $(JR)_d = x_n^d J_0 \oplus x_n^{d-1} J_1 \oplus \cdots \oplus x_n J_{d-1} \oplus J_d$ for any of $J = I'$ and $J = K[x_1, \ldots, x_{n-1}]$ (note that this holds also for $d \geq D$), the right hand side of (3.2.2) is equal to $\dim_K R_d - \dim_K (I'R)_d$, so that $\dim_K (I'R)_d = \dim_K I_d$. Here, it follows from [29, Theorem 5.2.6] that $\dim_K(\langle \mathrm{LM}_\prec(I) \rangle_R)_d = \dim_K(\langle \mathrm{LM}_\prec(I'R) \rangle_R)_d$. Since $\langle \mathrm{LM}_\prec(I'R) \rangle_R = \langle \mathrm{LM}_\prec(I') \rangle_R$, we finally obtain (3.2.1). $\square$

Lemma 3.2.2 (cf. [35, Lemma 2])**.** *Under the same setting as in Lemma 3.2.1, for each degree $d < D$, we have*

$$\mathrm{LM}_\prec(G)_d = \mathrm{LM}_\prec(G')_d. \tag{3.2.3}$$

Proof. This can be proved just by translating the proof of [35, Lemma 2]: Replace R', R, $\langle \boldsymbol{F}^h \rangle_{R'}$, $\langle \boldsymbol{F}^{\mathrm{top}} \rangle_R$, G_{hom}, and G_{top} in the proof by R, $K[x_1, \ldots, x_{n-1}]$, I, I', G, and G' in our case, respectively. $\square$

Lemma 3.2.3. *Under the same setting as in Lemma 3.2.1, we also suppose that $R/(I + \langle x_n \rangle)$ is Artinian (equivalently $K[x_1, \ldots, x_{n-1}]/I'$ is Artinian), namely $D < \infty$. Furthermore, we assume that $\langle \mathrm{LM}_\prec(I) \rangle_R$ is weakly reverse lexicographic. Then we have* $\mathrm{max.GB.deg}_\prec(I) \geq D$ *and* $\mathrm{max.GB.deg}_\prec(I') = D$.

Proof. Put $d := \mathrm{max.GB.deg}_\prec(I)$. Assume for a contradiction that $d < D$. Since $\mathrm{Krull.dim}(R/(I + \langle x_n \rangle)) = 0$, it follows from Lemma 2.1.5 that there is an element g in G with $\mathrm{LM}_\prec(g) = x_{n-1}^{d'}$ for some $d' \leq d$. Then, any monomial $t \in R_{d'}$ with $t \succeq x_{n-1}^{d'}$ (namely t is any monomial in $K[x_1, \ldots, x_{n-1}]_{d'}$) belongs to the initial ideal $\mathrm{in}_\prec(I) = \langle \mathrm{LM}_\prec(I) \rangle_R = \langle \mathrm{LM}_\prec(G) \rangle_R$ by the weakly reverse lexicographicness. Here, by $d < D$ together with (3.2.3), we have $\langle \mathrm{LM}_\prec(G) \rangle_R = \langle \mathrm{LM}_\prec(G')_{\leq d} \rangle_R$. This implies that t also belongs to $\langle \mathrm{LM}_\prec(G') \rangle_{K[x_1,\ldots,x_{n-1}]}$, and hence $K[x_1, \ldots, x_{n-1}]_{d'}/I'_{d'} = 0$. Thus, we have $d \geq d' \geq d_{\mathrm{reg}}(I')$, a contradiction.

By the same argument as above, we can show that $\max.\mathrm{GB.deg}_{\prec}(I') = d_{\mathrm{reg}}(I')$ as follows. Note that $\max.\mathrm{GB.deg}_{\prec}(I') \leq d_{\mathrm{reg}}(I')$ follows from a general fact written in the second paragraph of Remark 3.2.4 below. Thus, we suppose to the contrary that $d := \max.\mathrm{GB.deg}_{\prec}(I') < D := d_{\mathrm{reg}}(I')$. Then, it follows from (3.2.3) that $\mathrm{LM}_{\prec}(G') = \mathrm{LM}_{\prec}(G)_{\leq d}$, and the initial ideal $\mathrm{in}_{\prec}(\langle G' \rangle) = \langle \mathrm{LM}_{\prec}(G') \rangle_{K[x_1,\ldots,x_{n-1}]}$ has the *weak reverse lexicographicness up to d*. Since there is an element g in G' with $\mathrm{LM}_{\prec}(g) = x_{n-1}^{d'}$ for some $d' \leq d$, any monomial in $K[x_1,\ldots,x_{n-1}]_{d'}$ belongs to $\langle \mathrm{LM}_{\prec}(G') \rangle_{K[x_1,\ldots,x_{n-1}]}$ and so $K[x_1,\ldots,x_{n-1}]_{d'}/I'_{d'} = 0$, which implies $d \geq d' \geq D$, a contradiction. $\square$

Remark 3.2.4. In Lemma 3.2.3, when $R/(I + \langle x_n \rangle)$ is Artinian (or equivalently $K[x_1,\ldots,x_{n-1}]/I'$ is Artinian), we can easily prove $\max.\mathrm{GB.deg}_{\prec}(I') = D$ if $\mathrm{in}_{\prec}(I') = \langle \mathrm{LM}_{\prec}(I') \rangle_{K[x_1,\ldots,x_{n-1}]}$ is a weakly reverse lexicographic ideal, assuming neither that $(f_1,\ldots,f_m,x_n)$ is cryptographic semi-regular nor that $\mathrm{in}_{\prec}(I) = \langle \mathrm{LM}_{\prec}(I) \rangle_R$ is weakly reverse lexicographic.

In general, for any homogeneous ideal $J \subset R$ with $\mathrm{Krull.dim}(R/J) = 0$, we have $\max.\mathrm{GB.deg}_{\prec}(J) \leq \mathrm{reg}(J) = d_{\mathrm{reg}}(J)$ by [5, Corollary 2.5] (see also [30, Proposition 4.15]), and it is straightforward that the equality holds if the initial ideal $\langle \mathrm{LM}_{\prec}(J) \rangle_R$ of J is a weakly reverse lexicographic.

Proof of Theorem 1.2: Put $D := d_{\mathrm{reg}}(I + \langle x_n \rangle_R) = d_{\mathrm{reg}}(I')$ and $D' := \widetilde{d}_{\mathrm{reg}}(I)$. From our assumption that $R/(I + \langle x_n \rangle)$ is Artinian, we have the inequality (1.2), i.e., $\max.\mathrm{GB.deg}(I) \leq \max\{D, D'\}$. The case where $\mathrm{Krull.dim}(R/I) = 0$ is clear from Remark 3.2.4, so we assume that $\mathrm{Krull.dim}(R/I) = 1$. Note that, for the reduced Gröbner basis G of I, the elements of $\mathrm{LM}_{\prec}(G)$ are the minimal generators of the monomial ideal $\langle \mathrm{LM}_{\prec}(I) \rangle_R = \langle \mathrm{LM}_{\prec}(G) \rangle_R$.

First we consider the case where $D' \geq D$, so that $\max\{D, D'\} = D'$. Put $d := \max.\mathrm{GB.deg}_{\prec}(I) \leq D'$. Then, it follows from Macaulay's basis theorem (cf. [33, Theorem 1.5.7]) that R_d/I_d has a K-linear basis of the form

$$\{t \in R_d : t \text{ is a monomial and } t \notin \langle \mathrm{LM}_{\prec}(I) \rangle_R = \langle \mathrm{LM}_{\prec}(G) \rangle_R\},$$

which is called the *standard monomial basis*. We set $r := \dim_K R_d/I_d$, and write this standard monomial basis as $\{t_1,\ldots,t_r\}$ with $t_1 \succ \cdots \succ t_r$. In the following, we prove that $\{t_1 x_n^s,\ldots,t_r x_n^s\}$ is a K-linear basis of R_{d+s}/I_{d+s} for any s with $s \geq 1$, from which we have $\dim_K R_d/I_d = \dim_K R_{d+s}/I_{d+s}$ for any positive integer s, so that $d \geq D'$ and therefore $d = D'$.

By Lemma 3.2.3, we have $d \geq D$, and thus it follows from Remark 2.1.12 that the multiplication-by-x_n^s map from R_d/I_d to R_{d+s}/I_{d+s} is surjective (since $R/(I + \langle x_n \rangle)$ is Artinian). Therefore the set $B_s := \{t_1 x_n^s,\ldots,t_r x_n^s\}$ generates R_{d+s}/I_{d+s} over K. Suppose to the contrary that B_s is not a basis. In this case, there exists an i such that $t_i x_n^s$ is divisible by $\mathrm{LM}_{\prec}(g)$ for some $g \in G$. Indeed, since the elements of B_s are linearly dependent over K, we obtain $f :=$

$a_1t_1x_n^s + \cdots + a_rt_rx_n^s \in I_{d+s}$ for some $(a_1, \ldots, a_r) \in K^r \smallsetminus \{(0, \ldots, 0)\}$. Thus, for the least i with $a_i \neq 0$, the leading monomial of f is $t_ix_n^s$, which is divisible by $\mathrm{LM}_\prec(g)$ for some $g \in G$. Putting $u_i = \mathrm{GCD}(t_i, \mathrm{LM}_\prec(g))$ and $s_i = t_i/u_i$, we have $u_is_ix_n^s = t_ix_n^s$. Since $t_i = s_iu_i$ is not divisible by $\mathrm{LM}_\prec(g)$ and since $u_i = \mathrm{GCD}(t_i, \mathrm{LM}_\prec(g))$, we can write $u_ix_n^{s'} = \mathrm{LM}_\prec(g)$ for some $s' \leq s$. (We note that $\deg s_i = d - \deg u_i \geq \deg \mathrm{LM}_\prec(g) - \deg u_i = s'$.) Note that $s' \geq 1$ since otherwise t_i is divisible by $\mathrm{LM}_\prec(g)$.

Take an arbitrary monomial s_i' such that $\deg s_i' = s'$ and s_i' divides s_i. Then, by $\mathrm{LM}_\prec(g) = u_ix_n^{s'} \preceq u_is_i'$ together with the weakly reverse lexicographicness of the initial ideal $\mathrm{in}_\prec(I) = \langle \mathrm{LM}_\prec(I)\rangle_R$, the monomial $s_i'u_i$ should belong to $\langle \mathrm{LM}_\prec(G)\rangle_R$. Moreover, since $s_i'u_i$ divides $t_i = s_iu_i$, the monomial $t_i = s_iu_i$ also belongs to $\langle \mathrm{LM}_\prec(G)\rangle_R$, which is a contradiction.

Finally, consider the remaining case, i.e., $D > D'$, so that $\max\{D, D'\} = D$. In this case, it follows from Lemma 3.2.3 that $d \geq D$, whence $d = D$. $\square$

4 Results on genericness with conjectures

Let n be a positive integer, and let $R = K[x_1, \ldots, x_n]$ be the polynomial ring of n variables over a field K. We may assume that K is infinite. Let m, $d_1, \ldots, d_m$ be positive integers. As in Section 2.3, let $V_{n,m,(d_1,\ldots,d_m)} = R_{d_1} \times \cdots \times R_{d_m}$, and naturally endow it with the Zariski topology. Recall Definition 2.3.1 for the definition of the genericness of a property for $\boldsymbol{F} = (f_1, \ldots, f_m) \in V_{n,m,(d_1,\ldots,d_m)}$. For $\boldsymbol{F} = (f_1, \ldots, f_m) \in V_{n,m,(d_1,\ldots,d_m)}$, we denote by $\langle \boldsymbol{F} \rangle_R$ (or by $\langle \boldsymbol{F} \rangle$ simply) the ideal generated by $f_1, \ldots, f_m$.

4.1 Restricted genericness and our conjectures

For n, m, $(d_1, \ldots, d_m)$, and a non-negative integer k, we set

$$
\begin{aligned}
\Lambda^{\dim=k}_{n,m,(d_1,\ldots,d_m)} &:= \{\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)} : \mathrm{Krull.dim}(R/\langle \boldsymbol{F} \rangle) = k\},\\
\Lambda^{\dim\geq k}_{n,m,(d_1,\ldots,d_m)} &:= \{\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)} : \mathrm{Krull.dim}(R/\langle \boldsymbol{F} \rangle) \geq k\},\\
\Lambda^{\mathrm{csr}}_{n,m,(d_1,\ldots,d_m)} &:= \{\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)} : \boldsymbol{F} \text{ is cryptographic semi-regular}\},\\
\Lambda^{\mathrm{gcsr}}_{n,m,(d_1,\ldots,d_m)} &:= \{\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)} : \boldsymbol{F} \text{ is generalized cryptographic semi-regular}\}.
\end{aligned}
$$

It follows from Lemma 2.2.12 that $\Lambda^{\mathrm{gcsr}}_{n,m,(d_1,\ldots,d_m)} \cap \Lambda^{\dim=0}_{n,m,(d_1,\ldots,d_m)} \subset \Lambda^{\mathrm{csr}}_{n,m,(d_1,\ldots,d_m)}$ and the equality holds if and only if $m \geq n$.

Here, as a variant of Conjecture 2.3.7, we start with raising the following conjecture (cf. [36, Conjecture 4.3.4] as its strict version):

Conjecture 4.1.1. Assume that K is an infinite field. For any n, m, and $(d_1, \ldots, d_m)$, the property that $\boldsymbol{F} = (f_1, \ldots, f_m) \in V_{n,m,(d_1,\ldots,d_m)}$ is generalized cryptographic semi-regular is generic.

We see that Conjectures 2.3.7 and 4.1.1 are equivalent to each other. For this, we first prove the following lemma:

Lemma 4.1.2. *For any n, m, and $(d_1, \ldots, d_m)$ with $m \geq n$, the property that $R/\langle f_1, \ldots, f_m \rangle$ is Artinian is generic.*

Proof. The case $m = n$ is well-known to hold: More strongly, $\Lambda^{\dim=0}_{n,n,(d_1,\ldots,d_n)}$ is a non-empty Zariski-open set of $V_{n,n,(d_1,\ldots,d_n)}$.

Assume $m > n$, fix n, m, and $(d_1, \ldots, d_m)$. Then, the preimage of $\Lambda^{\dim=0}_{n,n,(d_1,\ldots,d_n)}$ by a natural projection sending an element of $V_{n,m,(d_1,\ldots,d_m)}$ to the first n-entries is included in $\Lambda^{\dim=0}_{n,m,(d_1,\ldots,d_m)}$. Since π is continuous, the preimage is Zariski-open in $V_{n,m,(d_1,\ldots,d_m)}$, and it is clearly non-empty, whence the assertion holds. $\square$

Proposition 4.1.3. *Assume that K is an infinite field. For $m \geq n$, Conjecture 4.1.1 is equivalent to Fröberg's conjecture (Conjecture 2.3.7).*

Proof. Assume that there exists a non-empty Zariski-open set $U_1 \subset V_{n,m,(d_1,\ldots,d_m)}$ with $U_1 \subset \Lambda^{\mathrm{gcsr}}_{n,m,(d_1,\ldots,d_m)}$. By Lemma 4.1.2, there exists a non-empty Zariski-open set $U_2 \subset V_{n,m,(d_1,\ldots,d_m)}$ such that $U_2 \subset \Lambda^{\dim=0}_{n,m,(d_1,\ldots,d_m)}$. Since both U_1 and U_2 are dense, we have $\emptyset \neq U_1 \cap U_2 \subset \Lambda^{\mathrm{gcsr}}_{n,m,(d_1,\ldots,d_m)} \cap \Lambda^{\dim=0}_{n,m,(d_1,\ldots,d_m)} \subset \Lambda^{\mathrm{csr}}_{n,m,(d_1,\ldots,d_m)}$, whence Fröberg's conjecture holds for $(n, m, d_1, \ldots, d_m)$.

The converse is obvious by $\Lambda^{\mathrm{csr}}_{n,m,(d_1,\ldots,d_m)} \subset \Lambda^{\mathrm{gcsr}}_{n,m,(d_1,\ldots,d_m)}$ (cf. Lemma 2.2.12). $\square$

Despite of the equivalence between two conjectures, those cannot guarantee that $\Lambda^{\mathrm{gcsr}}_{n,m,(d_1,\ldots,d_m)} \cap \Lambda^{\dim\geq 1}_{n,m,(d_1,\ldots,d_m)}$ includes a non-empty open set in the subspace $\Lambda^{\dim\geq 1}_{n,m,(d_1,\ldots,d_m)}$ equipped with the induced topology. To discuss this, we define the *restricted genericness* as follows:

Definition 4.1.4 (Restricted genericness)**.** Let V' be a non-empty subset of $V_{n,m,(d_1,\ldots,d_m)}$. For given n, m, and $(d_1, \ldots, d_m)$, a property $\mathcal{P}$ for $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ is said to be *generic on* V' if it holds on a non-empty open set of V' with respect to the induced topology, namely there exists a non-empty Zariski open set U in $V_{n,m,(d_1,\ldots,d_m)}$ such that $\emptyset \neq U \cap V' \subset \{\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)} : \boldsymbol{F} \text{ satisfies } \mathcal{P}\}$.

Namely, it is important whether the property that $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ is generalized cryptographic semi-regular is *generic on* $\Lambda^{\dim\geq 1}_{n,m,(d_1,\ldots,d_m)}$ or not. We will also discuss the genericness on

$$Z_{\boldsymbol{p}} := \{\boldsymbol{H} \in V_{n,m,(d_1,\ldots,d_m)} : \boldsymbol{p} \in V_{\overline{K}}(\boldsymbol{H})\} \tag{4.1.1}$$

for each point $\boldsymbol{p} \in \mathbb{P}^{n-1}(K)$.

Remark 4.1.5. If K is an infinite field and if the property that $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ is generalized cryptographic semi-regular is *generic on* $\Lambda^{\dim\geq 1}_{n,m,(d_1,\ldots,d_m)}$, then 'almost' all sequences $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ such that $\mathrm{Krull.dim}(R/\langle \boldsymbol{F}\rangle) \geq 1$ are expected to be generalized cryptographic semi-regular.

As the first discussion about the restricted genericness of the generalized cryptographic semi-regularity on $\Lambda^{\dim\geq 1}_{n,m,(d_1,\ldots,d_m)}$, we consider the following sub-case, where $\mathrm{Krull.dim}(R/\langle \boldsymbol{F}\rangle) = 1$ and $\dim_K R/\langle \boldsymbol{F}\rangle_D = 1$ for $D = \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{F}\rangle) < \infty$. In this case, $\langle \boldsymbol{F}\rangle$ has a unique projective zero $\boldsymbol{p}$ in $\mathbb{P}^{n-1}(K)$. First we show that it suffices to consider the case of the point $\boldsymbol{o} := (0 : \cdots : 0 : 1)$ as $\boldsymbol{p}$, by the following lemma:

Lemma 4.1.6. *If the property that $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ is generalized cryptographic semi-regular is generic on $Z_{\boldsymbol{o}}$, then it is also generic on $Z_{\boldsymbol{p}}$ for any $\boldsymbol{p} \in \mathbb{P}^{n-1}(K)$.*

Proof. For each $\boldsymbol{p} = (p_1 : \ldots : p_n) \in \mathbb{P}^{n-1}(K)$, let $\phi_{\boldsymbol{p}}$ be an invertible projective linear transformation sending $\boldsymbol{p}$ to $\boldsymbol{o}$. We may assume $p_n \neq 0$ and so $p_n = 1$ without loss of generality, so that we can take $\phi_{\boldsymbol{p}}$ as

$$(x_1 : \cdots : x_n) \to (x_1 - p_1 x_n : \cdots : x_{n-1} - p_1 x_{n-1} : x_n).$$

Then, the induced polynomial map $\phi_{\boldsymbol{p}}^* : R \to R$ is an isomorphism substituting $x_i - p_i x_n$ to x_i in $f \in R$ for $1 \leq i \leq n-1$ and stabilizing x_n. Here, we consider a map on $V_{n,m,(d_1,\ldots,d_m)}$ defined by

$$\Phi_{\boldsymbol{p}} : V_{n,m,(d_1,\ldots,d_m)} \to V_{n,m,(d_1,\ldots,d_m)} \ ; \ (f_1, \ldots, f_m) \mapsto (\phi_{\boldsymbol{p}}^*(f_1), \ldots, \phi_{\boldsymbol{p}}^*(f_m)).$$

It is straightforward that this is an invertible linear (and hence homeomorphic) map on $V_{n,m,(d_1,\ldots,d_m)}$. Moreover, it is easily checked that $\phi_{\boldsymbol{p}}^*$ does not change LM for any $h \in R$ with respect to the graded reverse lexicographical ordering $\prec$ such that $x_1 \succ \cdots \succ x_n$. Thus, for any subset S of R, we have $\mathrm{LM}(S) = \mathrm{LM}(\phi_{\boldsymbol{p}}^*(S))$, from which $\mathrm{in}_\prec(\langle \boldsymbol{F}\rangle) = \mathrm{in}_\prec(\langle \Phi_{\boldsymbol{p}}(\boldsymbol{F})\rangle)$ for $\boldsymbol{F}$ in $V_{n,m,(d_1,\ldots,d_m)}$. This also implies that $\mathrm{HS}_{R/\langle \boldsymbol{F}\rangle}(z) = \mathrm{HS}_{R/\mathrm{in}(\langle \boldsymbol{F}\rangle)}(z) = \mathrm{HS}_{R/\mathrm{in}(\langle \Phi_{\boldsymbol{p}}(\boldsymbol{F})\rangle)}(z) = \mathrm{HS}_{R/\langle \Phi_{\boldsymbol{p}}(\boldsymbol{F})\rangle}(z)$.

From our assumption, we have a non-empty Zariski-open set $U_{\boldsymbol{o}}$ of $V_{n,m,(d_1,\ldots,d_m)}$ intersecting with $Z_{\boldsymbol{o}}$ such that $U_{\boldsymbol{o}} \cap Z_{\boldsymbol{o}} \subset \Lambda^{\mathrm{gcsr}}_{n,m,(d_1,\ldots,d_m)} \cap Z_{\boldsymbol{o}}$. Then $\Phi_{\boldsymbol{p}}(U_{\boldsymbol{o}})$ is a non-empty Zariski-open set of $V_{n,m,(d_1,\ldots,d_m)}$ intersecting with $\Phi_{\boldsymbol{p}}(Z_{\boldsymbol{o}}) = Z_{\boldsymbol{p}}$. Moreover, as any $\boldsymbol{F} \in U_{\boldsymbol{o}} \cap Z_{\boldsymbol{o}}$ is generalized cryptographic semi-regular, $\Phi_{\boldsymbol{p}}(\boldsymbol{F})$ is also generalized cryptographic semi-regular, as $\mathrm{HS}_{R/\langle \boldsymbol{F}\rangle}(z) = \mathrm{HS}_{R/\langle \Phi_{\boldsymbol{p}}(\boldsymbol{F})\rangle}(z)$. Thus, letting $U_{\boldsymbol{p}} = \Phi_{\boldsymbol{p}}(U_{\boldsymbol{o}})$, we have $U_{\boldsymbol{p}} \cap Z_{\boldsymbol{p}} \subset \Lambda^{\mathrm{gcsr}}_{n,m,(d_1,\ldots,d_m)} \cap Z_{\boldsymbol{p}}$. $\square$

Note that

$$Z_{\boldsymbol{o}} = \{(f_1, \ldots, f_m) \in V_{n,m,(d_1,\ldots,d_m)} : \text{the } x_n^{d_j}\text{-coefficient of } f_j \text{ is zero } (1 \leq \forall j \leq m)\}.$$

Definition 4.1.7 (Generic sequence for $Z_{\boldsymbol{o}}$)**.** For checking the genericness of the generalized cryptographic semi-regularity of a sequence in $Z_{\boldsymbol{o}}$, we introduce a *generic sequence* $(g_1, \ldots, g_m)$ for $Z_{\boldsymbol{o}}$ with $\deg(g_j) = d_j$ as follows: Let Ω be an extension field of K, and let $\mathcal{C} := \{c_{j,t} : 1 \leq j \leq m,\ t \in \mathcal{T}_{d_j} \smallsetminus \{x_n^{d_j}\}\}$ be a set of elements in Ω that are algebraically independent over the prime field K_0 of K. Then we set $L := K(\mathcal{C})$ and $L_0 := K_0(\mathcal{C})$. Putting $g_j = \sum_{t \in \mathcal{T}_{d_j} \smallsetminus \{x_n^{d_j}\}} c_{j,t} t$ for each $1 \leq j \leq m$, we call a sequence $\boldsymbol{G} = (g_1, \ldots, d_m)$ in $K_0[\mathcal{C}][X] \subset L_0[X] \subset L[X]$ a generic sequence for $Z_{\boldsymbol{o}}$. We also set $\mathcal{F}_j := \left\{ \sum_{t \in \mathcal{T}_{d_j} \smallsetminus \{x_n^{d_j}\}} a_{j,t} t : a_{j,t} \in K \right\}$ for $1 \leq j \leq m$, so that $Z_{\boldsymbol{o}} = \mathcal{F}_{d_1} \times \cdots \times \mathcal{F}_{d_m}$.

Then we state an analogue of Lemma 2.3.2:

Lemma 4.1.8. *For any $\boldsymbol{F} = (f_1, \ldots, f_m) \in Z_{\boldsymbol{o}} \subset V_{n,m,(d_1,\ldots,d_m)}$, we have the coefficient-wise inequality*

$$\mathrm{HS}_{R/\langle f_1,\ldots,f_m \rangle}(z) = \mathrm{HS}_{R_L/\langle f_1,\ldots,f_m \rangle_{R_L}} \geq \mathrm{HS}_{R_L/\langle g_1,\ldots,g_m \rangle_{R_L}}(z), \tag{4.1.2}$$

where $(g_1, \ldots, g_m)$ is the generic sequence for $Z_{\boldsymbol{o}}$ defined in Definition 4.1.7, and where $R_L = L[X]$ with $L = K(\mathcal{C})$.

Proof. This can be proved similarly to Lemma 2.3.2, see [23, Lemma 1] for a proof by Fröberg. □

Proposition 4.1.9. *Assume that K is an infinite field. Then, for given n, m, and $(d_1, \ldots, d_m)$, the following conditions (1), (2), and (3) are equivalent to each other:*

(1) There exists a generalized cryptographic semi-regular sequence $\boldsymbol{F} \in Z_{\boldsymbol{o}}$ with $D := \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{F} \rangle) < \infty$ such that $\dim_K(R/\langle \boldsymbol{F} \rangle)_D = 1$.

(2) The property that $\boldsymbol{F} \in Z_{\boldsymbol{o}}$ is generalized cryptographic semi-regular with $D = \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{F} \rangle) < \infty$ satisfying $\dim_K(R/\langle \boldsymbol{F} \rangle)_D = 1$ is generic on $Z_{\boldsymbol{o}}$ (in terms of Definition 4.1.4).

(3) For $\mathcal{C}$, L, R_L, and $\boldsymbol{G} = (g_1, \ldots, g_m)$ as in Definition 4.1.7, we have

$$\mathrm{HS}_{R_L/\langle g_1,\ldots,g_m \rangle}(z) \equiv \frac{\prod_{j=1}^{m}(1 - z^{d_j})}{(1-z)^n} \pmod{z^{D'}} \tag{4.1.3}$$

and $\dim_L(R_L/\langle \boldsymbol{G} \rangle)_{D'} = 1$ for $D' = \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{G} \rangle) < \infty$.

Proof. The implication (2) ⇒ (1) is obvious.

To see (1) ⇒ (3), by Lemmas 2.2.6 and 4.1.8, we have the same inequalities as in (2.3.3). Since $\boldsymbol{F}$ is generalized cryptographic semi-regular (i.e., $\widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{F} \rangle)$-regular), $\boldsymbol{G}$ is also $\widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{F} \rangle)$-regular. Moreover, since $\mathrm{Krull.dim}(R_L/\langle \boldsymbol{G} \rangle) \geq 1$, the inequality $1 = \mathrm{HF}_{R/\langle \boldsymbol{F}\rangle}(d) \geq \mathrm{HF}_{R_L/\langle \boldsymbol{G}\rangle}(d)$ becomes an equality for any $d \geq D$, so that $\mathrm{HS}_{R/\langle \boldsymbol{F}\rangle}(z) = \mathrm{HS}_{R_L/\langle \boldsymbol{G}\rangle}(z)$ and $\widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{G}\rangle) = D < \infty$.

The implication (3) $\Rightarrow$ (2) follows from Theorem 2.3.4. $\square$

From Proposition 4.1.9 and Lemma 4.1.6, if there exists a generalized cryptographic semi-regular sequence $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ with $\dim_K(R/\langle \boldsymbol{F}\rangle)_D = 1$ for $D = \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{F}\rangle) < \infty$, it follows that $Z_{\boldsymbol{p}} \cap \Lambda^{\mathrm{gcsr}}_{n,m,(d_1,\ldots,d_m)}$ includes a non-empty Zariski-open set in $Z_{\boldsymbol{p}}$ equipped with the induced topology for any $\boldsymbol{p} \in \mathbb{P}^{n-1}(K)$, which implies the following conjecture:

Conjecture 4.1.10 (cf. [36, Conjecture 4.3.4])**.** Assume that K is an infinite field. For any n, m, and $(d_1, \ldots, d_m)$ and for any $\boldsymbol{p} \in \mathbb{P}^{n-1}(K)$, the property that $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ is generalized cryptographic semi-regular is generic on $Z_{\boldsymbol{p}}$.

Let us raise a more general conjecture:

Conjecture 4.1.11. Assume that K is an infinite field. For any n, m, and $(d_1, \ldots, d_m)$, the property that $\boldsymbol{F} \in V_{n,m,(d_1,\ldots,d_m)}$ is generalized cryptographic semi-regular is generic on $V^{\dim\geq 1}_{n,m,(d_1,\ldots,d_m)}$.

4.2 Genericness on cryptographic semi-regularity and weakly reverse lexicographicness

In this subsection, let K be an infinite field. Here we give a proof of Theorem 1.5 and Theorem 1.6 using the notion of "parametric Gröbner basis" for the ideal generated by a "generic sequence" $\boldsymbol{G} = (g_1, \ldots, g_m)$ under the setting in Section 2.3 and Definition 4.1.7. We fix a monomial ordering $\prec$ on the set $\mathcal{T}$ of monomials in $X = \{x_1, \ldots, x_n\}$ as the graded reverse lexicographical ordering with $x_1 \succ x_2 \succ \cdots \succ x_n$. As additional notations, we simply write $\mathrm{LM}(*)$ and $\mathrm{in}(*)$ for $\mathrm{LM}_{\prec}(*)$ and $\mathrm{in}_{\prec}(*)$, respectively, and we define the following:

For each element $\boldsymbol{a} \in \mathcal{A}_{(d_1,\ldots,d_m)} := \mathbb{A}^N(K) = \prod_{j=1}^{m} \mathbb{A}^{N_j}(K)$ with $N_j = \binom{n+d_j-1}{d_j}$, we define a map $\pi_{\boldsymbol{a}}$ from $K[\mathcal{C}][X]$ to $K[X]$ by substituting each $c_{j,t}$ with $\boldsymbol{a}_{j,t}$. The map $\pi_{\boldsymbol{a}}$ is generalized to a map from $K[\mathcal{C}][X]^m$ to $K[X]^m$, for which we use the same symbol. Thus, for a sequence $\boldsymbol{F}$ in $V_{n,m,(d_1,\ldots,d_m)}$, there exists $\boldsymbol{a}$ in $\mathcal{A}$ uniquely such that $\boldsymbol{F} = \pi_{\boldsymbol{a}}(\boldsymbol{G})$, where $\boldsymbol{G}$ is a generic sequence in terms of Fröberg, see the second paragraph of Section 2.3 for the definition. We also note that, for a polynomial h in $L[X]$ with $L = K(\mathcal{C})$, if $\pi_{\boldsymbol{a}}$ does not annihilate any of the denominators of its coefficients, $\pi_{\boldsymbol{a}}(h)$ can be defined. Then, for each sequence $\boldsymbol{F}$ in $V_{n,m,(d_1,\ldots,d_m)}$, let $\mathrm{GB}(\boldsymbol{F})$ denote the reduced Gröbner basis of the ideal $\langle \boldsymbol{F}\rangle_R$ with respect to $\prec$. Moreover, for the generic sequence $\boldsymbol{G}$, let $\mathrm{GB}(\boldsymbol{G})$ denote the reduced Gröbner basis of the ideal $\langle \boldsymbol{G}\rangle_{L_0[X]}$ with respect to $\prec$. Then, it follows from $\boldsymbol{G} \subset L_0[X] \subset L[X]$ that $\mathrm{GB}(\boldsymbol{G})$ is also the reduced Gröbner basis of the ideal $\langle \boldsymbol{G} \rangle_{L[X]}$ with respect to $\prec$. Therefore, in the following, for deciding the shape of $\mathrm{GB}(\boldsymbol{G})$, we assume that all elements in $\mathcal{C}$ are algebraically independent over K. (See Remark 2.3.3.) Recall that $\mathrm{LM}(\mathrm{GB}(\boldsymbol{F}))$ forms the minimal generating set of the initial ideal $\mathrm{in}(\langle \boldsymbol{F} \rangle_R) := \langle \mathrm{LM}(\langle \boldsymbol{F} \rangle_R) \rangle_R$. We simply write $\langle \boldsymbol{G} \rangle$ and $\langle \boldsymbol{F} \rangle$ for $\langle \boldsymbol{G} \rangle_{L[X]}$ and $\langle \boldsymbol{F} \rangle_{K[X]}$, respectively.

Theorem 4.2.1. *If there exists a cryptographic semi-regular sequence $\boldsymbol{F}$ in $V_{n,m,(d_1,\ldots,d_m)}$, that is, $\boldsymbol{F} = \pi_{\boldsymbol{a}}(\boldsymbol{G})$ for some element $\boldsymbol{a} \in \mathcal{A}_{(d_1,\ldots,d_m)}$, such that $\mathrm{in}(\langle \boldsymbol{F} \rangle)$ is weakly reverse lexicographic, then we have $\mathrm{GB}(\boldsymbol{F}) = \pi_{\boldsymbol{a}}(\mathrm{GB}(\boldsymbol{G}))$ and $\mathrm{LM}(\mathrm{GB}(\boldsymbol{F})) = \mathrm{LM}(\mathrm{GB}(\boldsymbol{G}))$. Thus, $\mathrm{in}(\langle \boldsymbol{G} \rangle)$ is also weakly reverse lexicographic. Then Theorem 2.3.4 can be applied directly to $\boldsymbol{G}$, from which it follows that there exists a non-empty Zariski-open set $U_{\mathcal{A}}$ of $\mathcal{A}_{(d_1,\ldots,d_m)}$ such that $\mathrm{LM}(\mathrm{GB}(\boldsymbol{F})) = \mathrm{LM}(\mathrm{GB}(\pi_{\boldsymbol{b}}(\boldsymbol{G})))$ and $\mathrm{in}(\langle \pi_{\boldsymbol{b}}(\boldsymbol{G}) \rangle)$ is weakly reverse lexicographic for any $\boldsymbol{b} \in U_{\mathcal{A}}$.*

We remark that, for each homogeneous ideal I, since the set of the leading monomials of the reduced Gröbner basis of I forms the minimal generating set of the initial ideal $\mathrm{in}(I)$, the property of "weakly reverse lexicographicness" depends on the leading monomials of the reduced Gröbner basis. Considering the subset U of $V_{n,m,(d_1,\ldots,d_m)}$ corresponding to $U_{\mathcal{A}}$ in Theorem 4.2.1, we have the following:

Corollary 4.2.2. *If there exists a cryptographic semi-regular sequence $\boldsymbol{F}$ in $V_{n,m,(d_1,\ldots,d_m)}$ such that $\mathrm{in}(\langle \boldsymbol{F} \rangle)$ is weakly reverse lexicographic, then there also exists a non-empty Zariski-open set U of $V_{n,m,(d_1,\ldots,d_m)}$ such that $\boldsymbol{H}$ is cryptographic semi-regular and $\mathrm{in}(\langle \boldsymbol{H} \rangle)$ is weakly reverse lexicographic for any $\boldsymbol{H}$ in U. Namely, Fröberg's conjecture and Moreno-Socias' one are both true for $(n, m, d_1, \ldots, d_m)$.*

Proof of Theorem 4.2.1. It suffices to show the following by induction argument on a non-negative integer d:

$$\mathrm{GB}(\boldsymbol{F})_d = \pi_{\boldsymbol{a}}(\mathrm{GB}(\boldsymbol{G})_d) \text{ with } \mathrm{LM}(\mathrm{GB}(\boldsymbol{F})_d) = \mathrm{LM}(\mathrm{GB}(\boldsymbol{G})_d). \tag{4.2.1}$$

For $d = 0$, it is clear as $\mathrm{GB}(\boldsymbol{F})_0 = \mathrm{GB}(\boldsymbol{G})_0 = \emptyset$. Thus, we show that the condition (4.2.1) holds for $d > 0$ under the *assumption on induction* that it holds for $d-1$. Put $\mathrm{GB}(\boldsymbol{F})_d = \{h_1, \ldots, h_v\}$ and $\mathrm{GB}(\boldsymbol{G})_d = \{\bar{h}_1, \ldots, \bar{h}_{\bar{v}}\}$ with $v = \#(\mathrm{GB}(\boldsymbol{F})_d)$ and $\bar{v} = \#(\mathrm{GB}(\boldsymbol{G})_d)$, where we order $h_1, \ldots, h_v$ and $\bar{h}_1, \ldots, \bar{h}_{\bar{v}}$ so that $\mathrm{LM}(h_1) \succ \mathrm{LM}(h_2) \succ \cdots \succ \mathrm{LM}(h_v)$ and $\mathrm{LM}(\bar{h}_1) \succ \mathrm{LM}(\bar{h}_2) \succ \cdots \succ \mathrm{LM}(\bar{h}_{\bar{v}})$, respectively. By the induction hypothesis, we have $\mathrm{LM}(\mathrm{GB}(\boldsymbol{F})_{<d}) = \mathrm{LM}(\mathrm{GB}(\boldsymbol{G})_{<d})$ and so $\mathrm{LM}(\langle \boldsymbol{F} \rangle_{<d}) = \mathrm{LM}(\langle \boldsymbol{G} \rangle_{<d})$. We let $Y := \mathrm{LM}(\langle \boldsymbol{F} \rangle_{<d})$. In the following we show $v = \bar{v}$ and $h_i = \pi_{\boldsymbol{a}}(\bar{h}_i)$ with $\mathrm{LM}(h_i) = \mathrm{LM}(\bar{h}_i)$ for each i.

<u>**Claim 1**</u>: $\bar{v} = v$ holds.

Let $W := \{ts : s \in Y,\ t \in \mathcal{T}_{d-\deg(s)}\}$. By the theory of Gröbner basis, the following holds:

$$\mathrm{LM}(\langle \boldsymbol{G} \rangle_d) = W \cup \mathrm{LM}(\mathrm{GB}(\boldsymbol{G})_d), \quad \mathrm{LM}(\langle \boldsymbol{F} \rangle_d) = W \cup \mathrm{LM}(\mathrm{GB}(\boldsymbol{F})_d)$$

Moreover, $\mathrm{LM}(\langle \boldsymbol{G}\rangle_d)$ (resp. $\mathrm{LM}(\langle \boldsymbol{F}\rangle_d)$) forms a linear basis of $\mathrm{in}(\langle \boldsymbol{G}\rangle)_d$ (resp. $\mathrm{in}(\langle \boldsymbol{F}\rangle)_d$), respectively. Here we note $\mathrm{HS}_{K[X]/\langle \boldsymbol{F}\rangle}(z) = \mathrm{HS}_{K[X]/\mathrm{in}(\langle \boldsymbol{F}\rangle)}(z)$ and $\mathrm{HS}_{L[X]/\langle \boldsymbol{G}\rangle}(z) = \mathrm{HS}_{L[X]/\mathrm{in}(\langle \boldsymbol{G}\rangle)}(z)$.

On the other hand, as $\boldsymbol{F}$ is cryptographic semi-regular, so is $\boldsymbol{G}$ by (1) $\Rightarrow$ (4) of Proposition 2.3.6, whence $\mathrm{HS}_{K[X]/\langle \boldsymbol{F}\rangle}(z) = \mathrm{HS}_{L[X]/\langle \boldsymbol{G}\rangle}(z)$. Therefore, their d-th coefficients, namely $\dim_K K[X]_d/\langle \boldsymbol{F}\rangle_d$ and $\dim_L L[X]_d/\langle \boldsymbol{G}\rangle_d$, coincide with each other, from which we have $v = \#(\mathrm{GB}(\boldsymbol{F})_d) = \#(\mathrm{GB}(\boldsymbol{G})_d) = \bar{v}$.

Claim 2: For each i, there is an element $\bar{\ell}_i$ in $\langle \boldsymbol{G}\rangle \cap K[\mathcal{C}][X]$ with $\mathrm{supp}(\bar{\ell}_i) \cap W = \emptyset$ such that $\pi_{\boldsymbol{a}}(\bar{\ell}_i)$ is defined and is a non-zero constant multiple of h_i. Here we denote by $\mathrm{supp}(h_i)$ the support of h_i, that is, the set of monomials appearing in h_i with non-zero coefficient.

For each i, as $h_i \in \langle \boldsymbol{F}\rangle$, there are polynomials $q_{i,1}, \ldots, q_{i,m} \in K[X]$ such that

$$h_i = \sum_{j=1}^{m} q_{i,j} f_j = \sum_{j=1}^{m} q_{i,j} \pi_{\boldsymbol{a}}(g_j) = \pi_{\boldsymbol{a}} \left(\sum_{j=1}^{m} q_{i,j} g_j \right).$$

Let $\ell_i := \sum_{j=1}^{m} q_{i,j} g_j$ for each i. Then ℓ_i belongs to $K[\mathcal{C}][X] \cap \langle \boldsymbol{G}\rangle$ as $g_i \in K[\mathcal{C}][X]$. Moreover, as $\mathrm{LM}(\pi_{\boldsymbol{a}}(\ell_i)) = \mathrm{LM}(h_i)$, we have $\mathrm{LM}(\ell_i) \succeq \mathrm{LM}(h_i)$. There are many choices on $q_{i,j}$, and ℓ_i may have some monomial in W with non-zero coefficient. Thus, to eliminate such a monomial in W from ℓ_i, we apply the following procedure to ℓ_i, which outputs a desired $\tilde{\ell}_i$:

(Step 1) Check if ℓ_i has no monomial in W, that is, $\mathrm{supp}(\ell_i) \cap W = \emptyset$ or not:

(Step 2: Case where $\mathrm{supp}(\ell_i) \cap W = \emptyset$)
We set $\tilde{\ell}_i := \ell_i$, and quite the procedure.

(Step 3: Case where $\mathrm{supp}(\ell_i) \cap W \neq \emptyset$)
Let t be the greatest monomial in $\mathrm{supp}(\ell_i) \cap W$ with respect to $\prec$. Then we compute ℓ_i' from ℓ_i by the following monomial reduction, by which the term with the monomial t is eliminated:

$$\ell_i' = \ell_i - b_t(\ell_i) \frac{t}{\mathrm{LM}(\bar{p})} \bar{p},$$

where $b_t(\ell_i)$ is the coefficient of t in ℓ_i and t is divided by $\mathrm{LM}(\bar{p})$ of some $\bar{p} \in \mathrm{GB}(\boldsymbol{G})_{<d}$. As $\pi_{\boldsymbol{a}}(\ell_i)$ is defined, $\pi_{\boldsymbol{a}}(b_t(\ell_i))$ is also defined. Moreover, by the assumption of induction, $\pi_{\boldsymbol{a}}(\bar{p})$ is defined and $\mathrm{LM}(\pi_{\boldsymbol{a}}(\bar{p})) = \mathrm{LM}(\bar{p})$. Therefore, $\pi_{\boldsymbol{a}}(\ell_i')$ is defined and we have the following image of the monomial reduction:

$$\pi_{\boldsymbol{a}}(\ell_i') = \pi_{\boldsymbol{a}}(\ell_i) - \pi_{\boldsymbol{a}}(b_t(\ell_i)) \frac{t}{\mathrm{LM}(\bar{p})} \pi_{\boldsymbol{a}}(\bar{p}) = h_i - \pi_{\boldsymbol{a}}(b_t(\ell_i)) \frac{t}{\mathrm{LM}(\pi_{\boldsymbol{a}}(\bar{p}))} \pi_{\boldsymbol{a}}(\bar{p})$$

As h_i is an element in $\mathrm{GB}(\boldsymbol{F})_d$, we have $\mathrm{supp}(\pi_{\boldsymbol{a}}(\ell_i)) \cap W = \mathrm{supp}(h_i) \cap W = \emptyset$, from which $\pi_{\boldsymbol{a}}(b_t(\ell_i)) = 0$ and $\pi_{\boldsymbol{a}}(\ell_i') = h_i$. In this case, if $\mathrm{supp}(\ell_i') \cap W$ is still not empty, its greatest monomial is smaller than t. Replacing ℓ_i with ℓ'_i, we go back to Step 1.

Since W is a finite set and the greatest monomial of $\mathrm{supp}(\ell_i)\cap W$ is decreasing, the procedure terminates in finitely many steps. Then, we convert the output polynomial $\tilde{\ell}_i$ to a polynomial $\bar{\ell}_i$ in $K[\mathcal{C}][X]$ by multiplying the least common multiple (LCM) of the denominators of coefficients of $\tilde{\ell}_i$ to itself. Then $\bar{\ell}_i$ satisfies the condition of Claim 2.

Claim 3: For each i, the image $\pi_{\boldsymbol{a}}(\bar{h}_i)$ is defined and it coincides with h_i.

Let $\mathrm{vect}(\bar{\ell}_i)$ be the *coefficient vector* of $\bar{\ell}_i$, where components are indexed by monomials in $\mathcal{T}_d$ in decreasing order. As $\bar{\ell}_i$ does not have any monomial in W, we consider a sub-vector $\mathrm{vect}_0(\bar{\ell}_i)$ which is obtained from $\mathrm{vect}(\bar{\ell}_i)$ by eliminating all coefficients of monomials in W. Let $t_1 \succ t_2 \succ \cdots \succ t_w$ are indices of vectors. (Thus, $\{t_1, \ldots, t_w\} = \mathcal{T}_d \setminus W$.) Since $\mathrm{in}(\langle \boldsymbol{F} \rangle)$ is weakly reverse lexicographic, it can be shown easily that $t_1 = \mathrm{LM}(h_1)$, $\ldots$, $t_v = \mathrm{LM}(h_v)$. Because, any monomial of degree d not smaller than $\mathrm{LM}(h_v)$ should belong to $W \cup \{\mathrm{LM}(h_1), \ldots, \mathrm{LM}(h_v)\}$, which implies that $\mathrm{LM}(h_1), \ldots, \mathrm{LM}(h_v)$ are top v monomials in $\mathcal{T}_d \setminus W$. Letting $\bar{\ell}_i = a_{i,t_1}t_1 + \cdots + a_{i,t_v}t_v + a_{i,t_{v+1}}t_{v+1} + \cdots + a_{i,t_w}t_w$ with $a_{i,t_j} \in K[\mathcal{C}]$ for each i, a sub-Macaulay matrix M over $K[\mathcal{C}]$ is constructed from vectors $\mathrm{vect}_0(\bar{\ell}_1), \ldots, \mathrm{vect}_0(\bar{\ell}_v)$ as follows:

$$M := \begin{pmatrix} a_{1,t_1} & \ldots & a_{1,t_v} & a_{1,t_{v+1}} & \ldots & a_{1,t_w} \\ \vdots & \ddots & \vdots & \vdots & & \vdots \\ a_{v,t_1} & \ldots & a_{v,t_v} & a_{v,t_{v+1}} & \ldots & a_{v,t_w} \end{pmatrix}$$

Then, we apply $\pi_{\boldsymbol{a}}$ naturally to M by $\pi_{\boldsymbol{a}}(M)_{i,t} = \pi_{\boldsymbol{a}}(M_{i,t})$ for each (i,t)-entry. As $\pi_{\boldsymbol{a}}(\bar{\ell}_i)$ is a non-zero constant multiple of h_i, we have

$$\pi_{\boldsymbol{a}}(M) = \begin{pmatrix} b_{1,t_1} & 0 & \ldots & 0 & b_{1,t_{v+1}} & \ldots & b_{1,t_w} \\ 0 & b_{2,t_2} & \ldots & 0 & b_{2,t_{v+1}} & \ldots & b_{2,t_w} \\ \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots \\ 0 & 0 & \ldots & b_{v,t_v} & b_{v,t_{v+1}} & \ldots & b_{v,t_w} \end{pmatrix},$$

where $b_{i,t_j} = \pi_{\boldsymbol{a}}(a_{i,t_j})$ for each i and j, and $b_{i,t_i} \neq 0$ for $1 \leq i \leq v$. Let M_0 be the square matrix consisting of the first v left columns of M and $\overline{M_0}$ the adjugate matrix of M_0. Namely we have

$$M_0 := \begin{pmatrix} a_{1,t_1} & \ldots & a_{1,t_v} \\ \vdots & \ddots & \vdots \\ a_{v,t_1} & \ldots & a_{v,t_v} \end{pmatrix},$$

and $\overline{M_0}$ is a matrix over $K[\mathcal{C}]$. Then, $\pi_{\boldsymbol{a}}(M_0)$ is a diagonal matrix with non-zero diagonal entries $b_{1,t_1}, \ldots, b_{v,t_v}$ and $\det(\pi_{\boldsymbol{a}}(M_0)) = \pi_{\boldsymbol{a}}(\det(M_0)) \neq 0$. Hence we have $\det(M_0) \neq 0$ as an element in $K[\mathcal{C}]$. Since $\overline{M_0}M_0 = \det(M_0)E_v$, where E_v denotes the identity matrix of degree v, we have

$$\overline{M_0}M = (\det(M_0)E_v, \overline{M_0}V_{v+1}, \ldots, \overline{M_0}V_w),$$

where V_j denotes the j-th column of M for each j.

For each i, let $\bar{h}_i'$ be a polynomial in $K[\mathcal{C}][X]$ converted from the i-th low of $\overline{M_0}M_0$. Then, since each $\bar{\ell}_i$ belongs to $\langle \boldsymbol{G} \rangle \cap K[\mathcal{C}][X]$ and the left-multiplication of $\overline{M}_0$ induces low-operations on M over $K[\mathcal{C}]$, each $\bar{h}_i'$ belongs to $\langle \boldsymbol{G} \rangle \cap K[\mathcal{C}][X]$. Moreover, we have $t_i = \mathrm{LM}(\bar{h}_i') \in \mathrm{in}(\langle \boldsymbol{G} \rangle)$ and $\mathrm{LC}(\bar{h}_i') = \det(M_0)$. Since t_i cannot be divided by any monomial in $\mathrm{LM}(\mathrm{GB}(\boldsymbol{F})_{<d}) = \mathrm{LM}(\mathrm{GB}(\boldsymbol{G}))_{<d})$, the monomials $t_1, \ldots, t_v$ should belong to the minimal generating set $\mathrm{LM}(\mathrm{GB}(\boldsymbol{G}))$ of $\mathrm{in}(\langle \boldsymbol{G} \rangle)$. Comparing the number $v = \#\mathrm{LM}(\boldsymbol{G}_d)$, we have $\{t_1, \ldots, t_v\} = \mathrm{LM}(\boldsymbol{G}_d)$.

Now let $\hat{h}_i = \frac{\bar{h}_i'}{\mathrm{LC}(\bar{h}_i')} = \frac{\bar{h}_i'}{\det(M_0)}$ for each i. Then, we have $\mathrm{LC}(\hat{h}_i) = 1$ for each i. Since $\mathrm{GB}(\boldsymbol{G})_{<d} \cup \{\hat{h}_1, \ldots, \hat{h}_v\}$ forms a set of inter-reduced elements, $\{\hat{h}_1, \ldots, \hat{h}_v\}$ coincides with $\mathrm{GB}(\boldsymbol{G})_d$, that is, $\bar{h}_i = \hat{h}_i$ for each i.

Finally we show $\pi_{\boldsymbol{a}}(\bar{h}_i) = h_i$ for each i. Since $\pi_{\boldsymbol{a}}(\det(M_0)) \neq 0$, the image $\pi_{\boldsymbol{a}}(\bar{h}_i)$ is defined for each i. Moreover, $\mathrm{LM}(\pi_{\boldsymbol{a}}(\bar{h}_i)) = t_i = \mathrm{LM}(h_i)$, $\mathrm{LC}(\pi_{\boldsymbol{a}}(\bar{h}_i)) = 1$, and $\bar{h}_i$ is *reduced* with respect to $\mathrm{GB}(\boldsymbol{G})_{<d}$ and t_j for $j \neq i$. This implies $\pi_{\boldsymbol{a}}(\bar{h}_i) = h_i$ for each i by the uniqueness of the reduced Gröbner basis with respect to a fixed monomial ordering. $\square$

Finally we consider a generalized cryptographic semi-regular sequence belonging to $Z_{\boldsymbol{o}}$, where $Z_{\boldsymbol{o}}$ is defined in Section 4.1 (see (4.1.1)). In this case, the set $\mathcal{C}$ of parameters is $\{c_{j,t} : 1 \leq j \leq m,\ t \in \mathcal{T}_{d_j} \smallsetminus \{x_n^{d_j}\}\}$. Moreover, we set $\mathcal{B}_d := \{(a_t)_{t \in \mathcal{T}_d \smallsetminus \{x_n^d\}} : a_t \in K\}$ for each positive integer d, and set $\mathcal{B}_{(d_1,\ldots,d_m)} := \prod_{j=1}^{m} \mathcal{B}_{d_j}$. Then, for each element $\boldsymbol{a}$ in $\mathcal{B}_{(d_1,\ldots,d_m)}$, we define a map $\pi_{\boldsymbol{a}}$ from $K[\mathcal{C}][X]$ to $K[X]$ by substituting each $c_{j,t}$ with $a_{j,t}$. The map $\pi_{\boldsymbol{a}}$ is generalized to a map from $K[\mathcal{C}][X]^m$ to $K[X]^m$, for which we use the same symbol. Let $\boldsymbol{G}$ be a generic sequence for $Z_{\boldsymbol{o}}$ defined in Definition 4.1.7. Under this setting, we can apply the same proof in Theorem 4.2.1 to obtain the following results corresponding to Theorem 4.2.1 and Corollary 4.2.2.

Proposition 4.2.3. *Let $\boldsymbol{G}$ be a generic sequence for $Z_{\boldsymbol{o}}$. If there exists a generalized cryptographic semi-regular sequence $\boldsymbol{F}$ in $Z_{\boldsymbol{o}}$, that is, $\boldsymbol{F} = \pi_{\boldsymbol{a}}(\boldsymbol{G})$ for some element $\boldsymbol{a} \in \mathcal{B}_{(d_1,\ldots,d_m)}$, such that $\dim_K(K[X]/\langle \boldsymbol{F} \rangle)_D = 1$ for $D = \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{F} \rangle) < \infty$ and $\mathrm{in}(\langle \boldsymbol{F} \rangle)$ is weakly reverse lexicographic, then $\mathrm{GB}(\boldsymbol{F}) = \pi_{\boldsymbol{a}}(\mathrm{GB}(\boldsymbol{G}))$ and $\mathrm{LM}(\mathrm{GB}(\boldsymbol{F})) = \mathrm{LM}(\mathrm{GB}(\boldsymbol{G}))$. In this case, $\boldsymbol{G}$ is generalized cryptographic semi-regular with $\dim_L(L[X]/\langle \boldsymbol{G} \rangle)_D = 1$ for $D = \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{G} \rangle) < \infty$ and $\mathrm{in}(\langle \boldsymbol{G} \rangle)$ is weakly reverse lexicographic. Moreover, in this case, there exists a non-empty Zariski-open set U of $Z_{\boldsymbol{o}}$ with respect to the induced topology such that $\boldsymbol{H}$ is generalized cryptographic semi-regular and $\mathrm{in}(\langle \boldsymbol{H} \rangle)$ is weakly reverse lexicographic for any $\boldsymbol{H}$ in U.*

Proof. By Proposition 4.1.9, we have

$$\mathrm{HS}_{K[X]/\langle \boldsymbol{F} \rangle}(z) \equiv \mathrm{HS}_{L[X]/\langle \boldsymbol{G} \rangle}(z) \equiv \frac{\prod_{j=1}^{m}(1-z^{d_j})}{(1-z)^n} \pmod{z^D},$$

where $D = \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{F} \rangle) = \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{G} \rangle)$ and $\dim_L(L[X]/\langle \boldsymbol{G} \rangle)_D = 1$. In this case, as $\boldsymbol{F}$ and $\boldsymbol{G}$ have $(0 : \cdots : 0 : 1)$ as their projective zero, x_n^D cannot belong neither to $\langle \boldsymbol{F} \rangle_D$ nor to $\langle \boldsymbol{G} \rangle_D$. This implies that $(K[X]/\langle \boldsymbol{F} \rangle)_D$ and $(L[X]/\langle \boldsymbol{G} \rangle)_D$ have a common linear basis $\{x_n^D\}$, which also implies that $\mathrm{max.GB.deg}(\langle \boldsymbol{F} \rangle) := \max\{\deg(h) : h \in \mathrm{GB}(\boldsymbol{F})\}$ and $\mathrm{max.GB.deg}(\langle \boldsymbol{G} \rangle)$ are both smaller than D. Thus, all arguments in the proof of Theorem 4.2.1 can be applied to the case of Proposition 4.2.3, from which we obtain $\mathrm{GB}(\boldsymbol{F}) = \pi_{\boldsymbol{a}}(\mathrm{GB}(\boldsymbol{G}))$ and $\mathrm{LM}(\mathrm{GB}(\boldsymbol{F})) = \mathrm{LM}(\mathrm{GB}(\boldsymbol{G}))$.

Then, by applying Theorem 2.3.4 to $\boldsymbol{G}$, there exists a non-empty Zariski-open set $U_{\mathcal{B}}$ of $\mathcal{B}_{(d_1,\ldots,d_m)}$ such that $\mathrm{LM}(\mathrm{GB}(\boldsymbol{F})) = \mathrm{LM}(\mathrm{GB}(\pi_{\boldsymbol{b}}(\boldsymbol{G})))$ for any $\boldsymbol{b}$ in U_B, that is, $\mathrm{in}(\langle \boldsymbol{F} \rangle) = \mathrm{in}(\langle \pi_{\boldsymbol{b}}(\boldsymbol{G}) \rangle))$. This implies that the subset U of $Z_{\boldsymbol{o}}$ corresponding to $U_{\mathcal{B}}$ is also a non-empty Zariski-open set such that $\boldsymbol{H}$ is generalized cryptographic semi-regular and $\mathrm{in}(\langle \boldsymbol{H} \rangle)$ is weakly reverse lexicographic for any $\boldsymbol{H} \in U_{\mathcal{B}}$. □

Proposition 4.2.4. *Suppose that the conditions of Proposition 4.2.3 are satisfied. Then, for any $\boldsymbol{p} \in \mathbb{P}^{n-1}(K)$, there exists a non-empty Zariski-open set $U_{\boldsymbol{p}}$ of $Z_{\boldsymbol{p}}$ with respect to the induced topology such that $\boldsymbol{H}$ is generalized cryptographic semi-regular and $\mathrm{in}(\langle \boldsymbol{H} \rangle)$ is weakly reverse lexicographic for any $\boldsymbol{H}$ in $U_{\boldsymbol{p}}$.*

Proof. We show the assertion by using the arguments in the proof of Lemma 4.1.6 Let $\boldsymbol{p} := (p_1 : \ldots : p_n)$ be a point in $\mathbb{P}^{n-1}(K)$. Without loss of generality, we may assume that $p_n \neq 0$ and moreover $p_n = 1$, and consider the following maps, where $\mathcal{V} := V_{n,m,(d_1,\ldots,d_m)}$:

$$\begin{aligned}
&\phi_{\boldsymbol{p}} : (a_1 : \cdots : a_{n-1} : a_n) \mapsto (a_1 - p_1 a_n : \cdots : a_{n-1} - p_{n-1} a_n : a_n),\\
&\phi_{\boldsymbol{p}}^* : K[X] \ni h(x_1, x_2, \ldots, x_n) \mapsto h(x_1 - p_1 x_n, \ldots, x_{n-1} - p_{n-1} x_n, x_n) \in K[X],\\
&\Phi_{\boldsymbol{p}} : \mathcal{V} \in (h_1, \ldots, h_m) \mapsto (\phi_{\boldsymbol{p}}^*(h_1), \ldots, \phi_{\boldsymbol{p}}^*(h_m)) \in \mathcal{V}.
\end{aligned}$$

Then $\phi_{\boldsymbol{p}}^*$ is a ring homomorphism and $\Phi_{\boldsymbol{p}}$ is an invertible linear map on $\mathcal{V}$ and its restriction on $Z_{\boldsymbol{o}}$ induces a bijection between $Z_{\boldsymbol{o}}$ and $Z_{\boldsymbol{p}}$. We also let $\prec$ be the graded reverse lexicographical ordering with $x_1 \succ x_2 \succ \cdots \succ x_n$.

Now we suppose that the conditions of Proposition 4.2.3 hold. Then the generic sequence $\boldsymbol{G}$ for $Z_{\boldsymbol{o}}$ is generalized cryptographic semi-regular sequence with $\dim_L(L[X]/\langle \boldsymbol{G} \rangle)_D = 1$ for $D = \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{G} \rangle) < \infty$ and $\mathrm{in}(\langle \boldsymbol{G} \rangle)$ is weakly reverse lexicographic, and there is a non-empty Zariski-open set $U_{\boldsymbol{o}}$ in $Z_{\boldsymbol{o}}$ such that, for any $\boldsymbol{H}$ in $U_{\boldsymbol{o}}$, $\boldsymbol{H}$ is generalized cryptographic semi-regular and $\mathrm{in}(\langle \boldsymbol{H} \rangle)$ is weakly reverse lexicographic.

Meanwhile, since $\phi_{\boldsymbol{p}}^*$ does not change LM for any polynomial in $L[X]$ and $\mathrm{in}(\langle \boldsymbol{F} \rangle) = \mathrm{in}(\langle \Phi_{\boldsymbol{p}}(\boldsymbol{F}) \rangle)$ for any $\boldsymbol{F}$ in $U_{\boldsymbol{o}}$, it can be shown easily that, for any $\boldsymbol{H}$ in $U_{\boldsymbol{o}}$, $\Phi_{\boldsymbol{p}}(\boldsymbol{H})$ is generalized cryptographic semi-regular and $\mathrm{in}(\langle \Phi_{\boldsymbol{p}}(\boldsymbol{H}) \rangle)$ is weakly reverse lexicographic. Let $U_{\boldsymbol{p}} = \Phi_{\boldsymbol{p}}(U_{\boldsymbol{o}})$. Then, $U_{\boldsymbol{p}}$ is also a non-empty Zariski-open set of $\Phi_{\boldsymbol{p}}(Z_{\boldsymbol{o}}) = Z_{\boldsymbol{p}}$ (see the proof of Lemma 4.1.6). Thus, $U_{\boldsymbol{p}}$ satisfies the assertion of Proposition 4.2.4. $\square$

Remark 4.2.5. It is clear that the converse of Proposition 4.2.4 also holds, that is, if there is a generalized cryptographic semi-regular sequence $\boldsymbol{H}$ in $Z_{\boldsymbol{p}}$ such that $\dim_K(K[X]/\langle \boldsymbol{H} \rangle)_D = 1$ for $D = \widetilde{d}_{\mathrm{reg}}(\langle \boldsymbol{H} \rangle) < \infty$ and $\mathrm{in}(\langle \boldsymbol{H} \rangle)$ is weakly reverse lexicographic, then the sequence $\boldsymbol{F} = (\Phi_{\boldsymbol{p}})^{-1}(\boldsymbol{H})$ belongs to $Z_{\boldsymbol{o}}$ and it satisfies the condition of Proposition 4.2.3.

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